Linköping University | Department of Computer and Information Science
Master's thesis, 30 ECTS | Datateknik
2026 | LIU-IDA/LITH-EX-A–26/062–SE

AI-Assisted Lesson Generation in Medical Education

– Design and Evaluation of a Generative AI Prototype for Educators

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AI-assisterad lektionsgenerering inom medicinsk utbildning
– Design och utvärdering av en generativ AI-prototyp för utbildare

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Abstract

Educators are facing heavy workloads, with potential consequences for both their well-being and the education system. As generative AI grows more capable, this raises the question of whether it can help reduce that burden. This project was focused on the generation of learning materials in Sectra's platform for medical education: Sectra Education Portal. The project was based on an existing prototype for lesson generation and aimed to understand educators' perceptions of the tool, the factors shaping those perceptions, and how to improve both through feature-level enhancements to the prototype. To answer these questions, a pre-study was conducted, consisting of a literature search on design principles for AI-powered systems, along with an interview round to gather user opinions of the original prototype. Then, enhancements to the prototype were designed and implemented, based on the findings from the pre-study. Finally, the original prototype and the extended version were compared in a user evaluation with 15 participants.

The study concluded that the enhancements positively affected the perceived usefulness, perceived ease of use, and trust in the prototype. The addition of an AI Chat stood out as an impactful enhancement for both perceived usefulness and ease of use. On the other hand, additions such as an AI Provenance Banner appeared to have no meaningful impact on the perceptions of study participants. Notably, participants reported an identical need to verify the tool's output between both prototypes, suggesting that perceived trust did not necessarily reduce need for verification.

Acknowledgments

I have a lot of people to thank for helping me carry out this project, which marks the end of my studies (for now): the Medical Education team at Sectra, for not only giving me this opportunity, but also helping with the development process, and, honestly, being great company along the way. The participants in this study, for taking the time to share their perspectives. My supervisor, Mohsen Asgari, for guiding and supporting me. My examiner, Aseel Berglund, for the constructive feedback and for being an inspiration over the last five years. My external supervisor, Emilia Ståhlbom, for putting up with me daily, for answering my countless questions, and for sharing her knowledge. My friends and family, for having my back and for being proud of me. Most importantly, I would like to thank Isak for being my biggest cheerleader and greatest inspiration.

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1 Introduction

Great advancements have been made over the last few years in the field of generative artificial intelligence (generative AI) [1]. Generative systems are now capable of producing data with a level of coherence that was previously out of reach, and these advancements, among other things, have expanded the possibilities of AI-assisted learning. This thesis investigates how learning materials for medical students can be produced using generative AI.

1.1 Motivation

Healthcare, and, by extension, the education of clinicians, is essential to society. Medical and healthcare education has several defining characteristics: unlike many other domains, education continues into professional practice, as evolving medical knowledge must be adopted and applied by practitioners [2]. Consequently, healthcare professionals engage in formal learning throughout their careers. It is also competence-oriented [2]: professionals must not only retain large amounts of information but also apply knowledge safely and effectively in real-world scenarios. Because errors in this domain can harm patients, the accuracy and reliability of educational materials are paramount. In short, high-quality medical education directly affects patient safety and public health.

However, producing and updating high-quality learning materials is resource-intensive and time-consuming for educators, who often face demanding workloads and limited time [3]. One potential way to address this challenge is to automate certain aspects of teaching, such as the creation of learning materials.

Education is one of the domains heavily affected by recent advances in generative AI [4]. The promises identified with generative AI in this domain include personalized learning and tutoring. At the same time, significant challenges have been identified, including ensuring ethical and responsible use, maintaining academic integrity, and mitigating risks such as plagiarism and misinformation [4]. Building on recent advancements in generative AI, this thesis investigates how such technologies can support the automated generation of learning materials.

Automating the generation of medical learning materials carries risks, such as hallucinations: false or fabricated content generated by AI [5]. Recent studies suggest that hallucinations cannot be fully eliminated in AI-generated content [6]. Given the direct implications of medical education for public health, any automatically generated materials must meet strict

requirements for factual accuracy and reliability, particularly in commercial contexts. Because of such risks, how educators perceive and place trust in AI-powered tools becomes central. Therefore, the focus of this project is educators' perceptions of AI-generated lessons rather than measuring the quality of the generated content.

This thesis is conducted in collaboration with Sectra and its Department of Medical Education, which develops the platform Sectra Education Portal. The platform is used by schools and teaching hospitals to support students' learning in the medical domain [7]. Among other functionalities, Sectra Education Portal enables digital lessons, which are slide-based modules that combine medical images, explanatory text, and questions in free-text, single-choice, or multiple-choice formats. This thesis explores the automated generation of such lessons using generative AI, with the aim of supporting educators. While focused on a medical domain, the results of this research may apply to other educational contexts as well, as similar challenges exist in other educational fields.

1.2 Aim

The main purpose of this thesis is to explore how AI can be used to generate digital lessons, with a focus on medical education. The thesis also examines how educators perceive the usefulness, ease of use, and trust of AI-generated medical lessons. To support this work, a prototype lesson-generation feature is implemented within the Sectra Education Portal platform to provide a concrete setting for generating and evaluating AI-produced learning materials. Specifically, the thesis compares two system designs for lesson generation.

1.3 Research questions

1. How can an AI lesson generator prototype for medical educators be extended through feature-level enhancements to improve its perceived usefulness, ease of use, and trust?
2. How do medical educators perceive the usefulness, ease of use, and trust of AI-generated lessons?
3. What factors influence medical educators' perceptions of usefulness, ease of use, and trust in AI-generated lessons?
4. How do feature-level enhancements to the prototype affect medical educators' perceived usefulness, ease of use, and trust?

1.4 Delimitations

This study is limited to lesson generation within the application Sectra Education Portal, and the results may therefore not generalize to other educational systems. The definition of a lesson within this context can be found in Section 2.2. Additionally, the user evaluation focuses on the perceptions of educators within the medical field. As a result, the study does not address learning outcomes from using the lessons, nor does it objectively assess the factual quality or accuracy of the generated content; the focus is on educators' subjective perceptions rather than independent measures of output quality. Since the implemented system is a prototype, user perceptions may not reflect those of a fully mature product.

2 Background

This chapter introduces the context in which the thesis is situated, starting with the company Sectra. This is followed by an introduction to the platform Sectra Education Portal and the lesson structure that this thesis is centered around. The typical users of Sectra Education Portal are described, before generative AI in education is given a brief overview.

2.1 Sectra

Sectra is a company based in Sweden, specializing in medical imaging IT, and cybersecurity [8]. Sectra's products include encryption solutions and the secure communications solution Tiger, used by Dutch ministries and the Swedish Defense, among others [9]. Furthermore, Sectra's medical products include an imaging platform for clinical contexts, surgery planning tools, and the platform Sectra Education Portal, where the current project resides [7]. Sectra Education Portal is further described below in Section 2.2.

2.2 Sectra Education Portal and lessons

The present project is conducted within the platform Sectra Education Portal, which is developed by Sectra. Sectra Education Portal is a web application used for medical education. Its primary users are educators responsible for creating instructional materials, while learners interact with the content as part of their training.

In addition to viewing medical images and cases, users can create lessons around these materials using Sectra's tools for medical image visualization and diagnostics. These lessons are typically structured as a sequence of cards containing images, explanatory text, multiple-choice questions, and free-text responses, see Figure 2.1. Furthermore, the lessons can serve as assessment tools, as they allow educators to include various types of questions that can be answered by students and subsequently reviewed and graded. Creating such lessons traditionally involves manually selecting relevant medical images from the digital library in Sectra Education Portal, structuring textual content, and formulating assessment questions.

Lessons may support different kinds of learning objectives as described in Bloom's Taxonomy, see Section 3.1. For example, a lesson may be used to present factual information to the learners, corresponding to lower-level learning, but may also present new medical cases

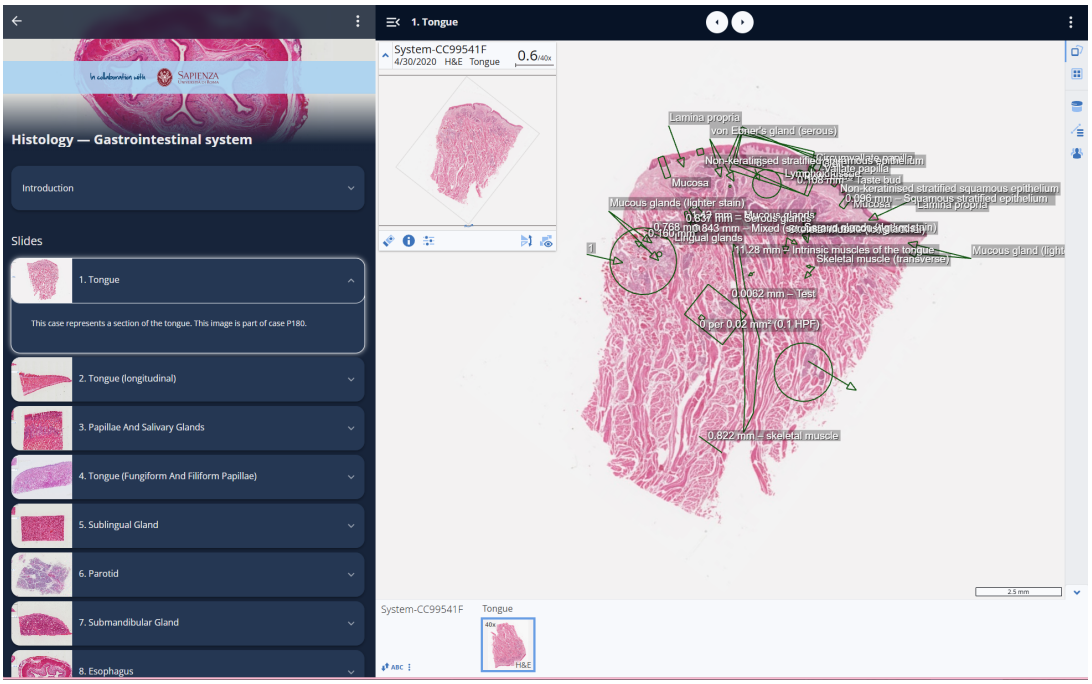


Figure 2.1: A lesson in Sectra Education Portal. The slides are listed on the left, while the image associated with the selected slide is displayed on the right. Although slides may contain different types of questions, the selected slide in this example consists of text only.

that require learners to apply previously acquired knowledge, corresponding to higher-level learning.

2.2.1 Users and usage context

At the time of writing, Sectra Education Portal is primarily used by educators and learners in universities across Europe, North and South America, and Oceania. The application primarily supports anatomy and radiology training. Training commonly involves medical images such as X-ray, CT, and MRI.

Lessons are typically created by Sectra personnel in collaboration with university educators, or by the university educators themselves. In some cases, the educators are near-peer teachers, i.e., more advanced students teaching those earlier in their training. Typical goals include creating high-quality educational materials, providing a practice environment, and assessing learners. However, commonly reported challenges include limited time for authoring the lessons.

2.2.2 User personas

To further illustrate the primary user group, two representative personas are described below. The personas are based on input from Sectra personnel and insights gathered during user interviews conducted as part of this study, which are described in Section 4.3.

Clinical educator: A faculty member responsible for teaching a university radiology course, alongside clinical practice. This educator uses Sectra Education Portal to assemble lesson materials and medical images in a clinically authentic environment. The teacher's goals include motivating students, providing standardized, high-quality materials, enabling remote practice, and assessing learners' knowledge and skills. One key challenge is finding appropriate

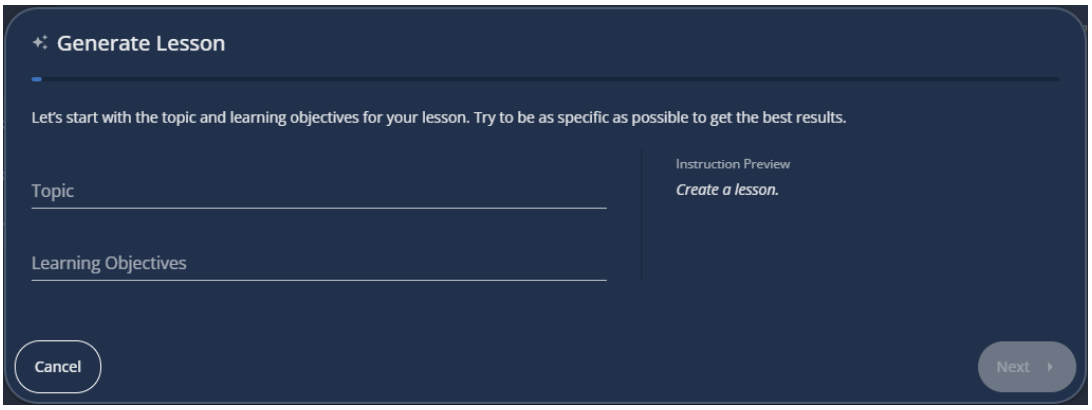


Figure 2.2: The first step of filling in the lesson generation prompt form in the original prototype. Topic and Learning objectives are obligatory fields.

images aligned with learning objectives – a task that requires domain expertise and significant time. Another challenge is limited time and heavy workloads, meaning that lesson authoring often competes with clinical and research responsibilities.

Customer success manager (CSM): A Sectra employee working in close relation to customers of Sectra Education Portal to help them use the platform effectively. The CSM assists teachers in creating lessons, with the goal of creating lessons that are useful to the teacher and meet their standards. In practice, this often means building the lessons on behalf of teachers: the teacher describes the content they want, and the CSM constructs it manually. As the Sectra Education Portal user base grows, demand for this support is outpacing capacity, making it increasingly important to find faster ways to produce high-quality lessons.

2.2.3 Existing lesson generator prototype

The existing lesson generator prototype was integrated into Sectra Education Portal. Users could initiate lesson generation by pressing a button. This opened up a form that guided them in constructing a prompt for the generative AI system. The full form is shown in Figure 2.2 to Figure 2.5, and contains the following fields for the user to fill in:

- Topic
- Learning objectives
- Type of lesson
- Level of education
- Number of slides (optional)
- Number of questions (optional)
- Additional context (optional)

The filled-in form was used to create a prompt, which was sent to an AI agent developed using Azure AI Foundry: Microsoft's cloud platform for building and deploying AI agents. The agent had access to a knowledge base containing medical images from the image library in Sectra Education Portal. The agent would then generate the entire lesson structure in a single response, which was presented to the user in an editable format, allowing manual modifications to the content.

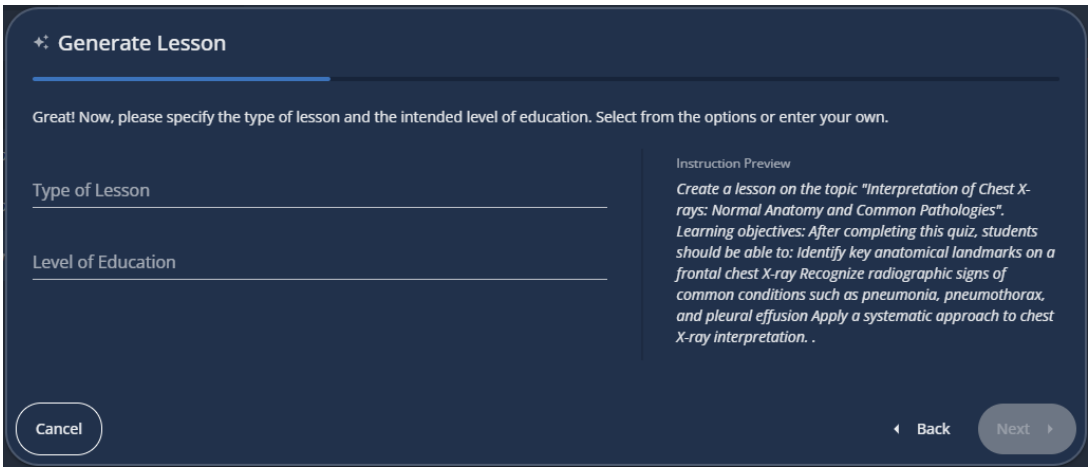


Figure 2.3: The second step of the lesson generator prompt form. The fields "Type of lesson" and "Level of Education" are drop-down menus with custom input possible. They are both mandatory fields.

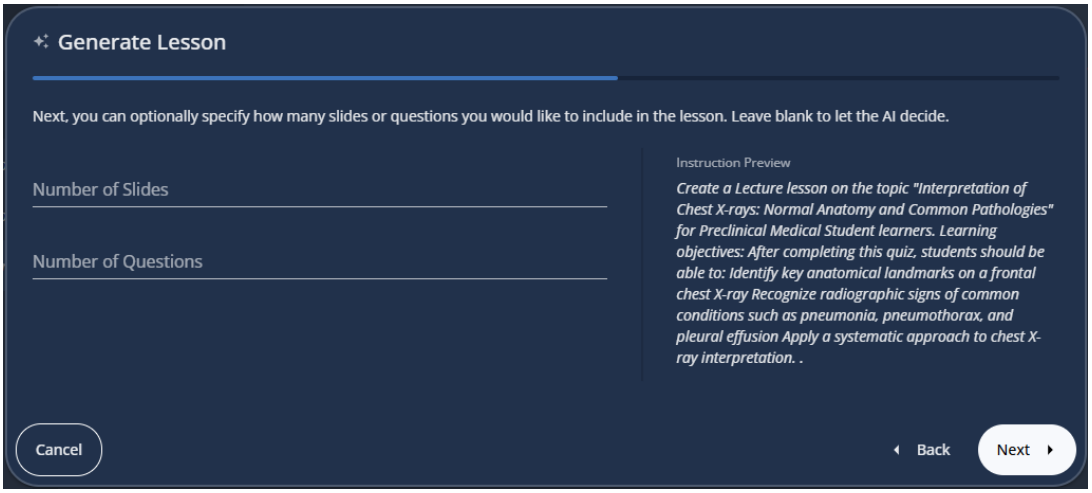


Figure 2.4: The third step of the lesson generator prompt form. Both fields are optional and may be left empty.

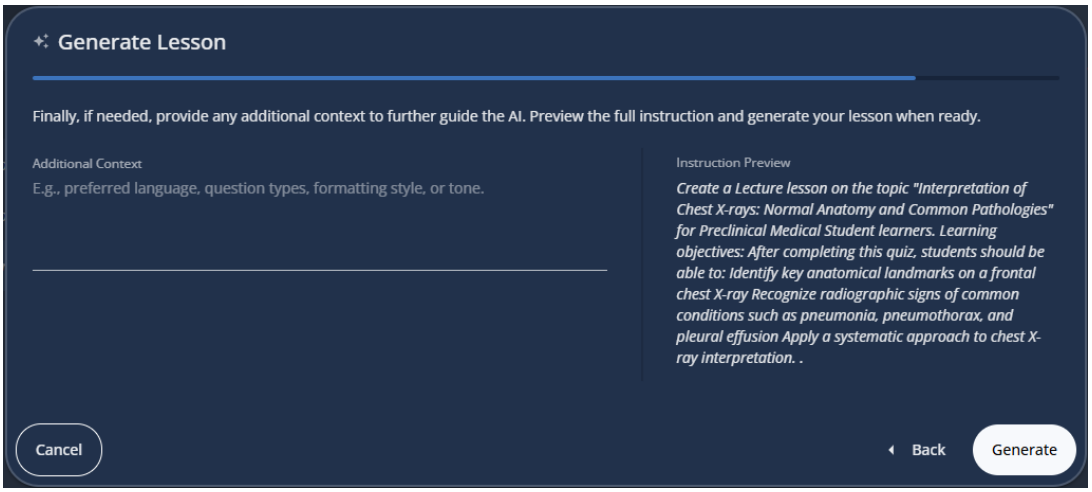


Figure 2.5: The fourth step of the lesson generator prompt form. Providing additional details is optional.

2.3 Generative AI in education

Generative AI refers to AI systems capable of creating new content, instead of retrieving fixed answers [10]. Due to the output being created rather than fetched, the output may vary even when prompting the system in the same way. Additionally, the way that prompts are written may affect the output significantly. As previously mentioned, generative AI is also capable of producing plausible but incorrect content: so-called hallucinations [5].

In education, generative AI has been used for tasks such as generating instructional content, including lessons and quizzes, as well as for personalizing content and providing automated learner feedback [11]. In medical education specifically, the potential benefits are significant, as producing high-quality learning materials is resource-intensive. Yet, the stakes are also high, given that inaccuracies in educational content may ultimately affect patient safety. In the context of this thesis, generative AI is particularly relevant as a tool for supporting educators in lesson generation. However, its use also introduces challenges, including the risk of propagating biases and concerns related to data privacy [12]. The state of generative AI in education is further explored in Section 3.5.2.

3 Theory

This chapter lays the groundwork for the thesis. It introduces Bloom's Taxonomy as a learning framework, outlines design principles for generative AI systems, summarizes technology-acceptance frameworks, and explains Retrieval-Augmented Generation (RAG). It concludes with an overview of related work.

3.1 Bloom's Taxonomy

As previously stated, the nature of the medical sector places strict demands on the education of practitioners. To structure learning objectives and educational design, Bloom's taxonomy is one of the most widely used educational frameworks [13] [14].

In the original Bloom's Taxonomy framework, learning objectives are divided into six classes called knowledge, comprehension, application, analysis, synthesis, and evaluation [13]. These categories are hierarchical, such that higher levels build upon lower ones — for example, knowledge acquisition precedes comprehension.

The knowledge class encompasses the ability to recall facts, methods, theoretical concepts, and such. The subsequent category, comprehension, refers to the ability to interpret information, translate knowledge between levels of abstraction, and extrapolate meaning. Building on this, the application category involves using learned concepts correctly in new situations without explicit guidance on which abstractions to apply. Analysis, the fourth category, entails breaking information into components, identifying relationships, and recognizing underlying principles. These abilities extend into the fifth class, synthesis, which involves combining elements into coherent products or generating new structures that go beyond the original material. Finally, evaluation concerns the ability to determine the value of ideas, methods, and solutions, based on implicit or explicit criteria [13].

3.2 Design principles

This section presents established design principles relevant to the design of generative AI systems. For clarity, the principles are organized here into six themes: human oversight and co-creation, usability and user interaction, system transparency, appropriate trust and reliance, responsible AI design, and integration into existing workflows.

3.2.1 Human oversight and co-creation

A central design principle for generative AI systems is that users should remain in control of the generation process while collaborating with the AI. Weisz et al. [15] emphasize that generative AI systems should not take complete control over a task, but rather work together with the user to support, rather than replace, human decision-making. This also applies to the revision of existing material, where the AI and the user should be able to co-create and iteratively co-edit content.

Similarly, Lin et al. [16] recommend allowing direct user modification of AI-generated output, while also letting the user iteratively improve the output with the help of AI. Such interaction patterns support co-creation by allowing users to refine individual elements of generated content rather than regenerating entire outputs.

Zamfirescu-Pereira et al. [17] further suggest that generative AI systems should avoid appearing overly human-like. When an AI system is presented as a social actor, users may feel constrained by social norms and hesitate to give direct instructions or critical feedback. Designing AI systems that clearly function as tools rather than social actors may therefore encourage users to interact more freely and maintain control over the generation process.

Together, these design recommendations emphasize that generative AI systems should support collaborative interaction patterns in which users remain active participants in the creation process.

3.2.2 Supporting user interaction and usability

While human oversight ensures that users remain in control of the generation process, users must also be supported in interacting effectively with generative AI systems. Effective interactions with generative AI often require users to interact through prompts, which can pose a barrier for non-expert users. Research by Zamfirescu-Pereira et al. [17] shows that prompting can be challenging, particularly when users lack an understanding of how their inputs influence system outputs. The authors therefore recommend providing the user with examples of successful prompts.

Another recommendation made by Zamfirescu-Pereira et al. [17] to give the user better prompting skills is to give them a greater understanding of cause-and-effect in the system. The authors say one way to achieve this could be to highlight prompt changes and their resulting outputs to the user. Lin et al. [16] propose design guidelines that help users provide more effective input, such as assisting users in improving their prompts or configuring generation parameters. The authors also state that users should be encouraged to give feedback to the system.

Weisz et al. [15] suggest that users should be given knowledge on how to prompt effectively, in order to co-create material with generative AI. The user should be guided to specify goals, constraints, and preferences for the desired output, rather than low-level implementation details. Generic aspects of the prompt, as well as controls specific to the context and use case, should be exposed to the user, for example, through sliders or other selection fields. By supporting co-creation, the system positions the user as an active collaborator rather than a passive recipient of AI-generated content.

In addition to these generative AI-specific guidelines, established usability principles should also be considered when designing generative AI systems. Jakob Nielsen's usability heuristics [18] highlight several relevant aspects. Nielsen says that how well the system matches the user's real world is an important factor in how usable the system is. This entails, among other things, that the language used in the system and the conceptual model of the system must both be familiar and natural. The author also highlights that the system status should be visible to the user to help them understand the effects of their actions.

3.2.3 User understanding and system transparency

Generative AI systems are non-deterministic in nature, meaning that the same input can yield different outputs. This variability can make it difficult for users to predict system behavior and to understand why certain outputs are produced. Weisz et al. [15] encourage designers to consider whether multiple outputs should be shown to the user, or if only one alternative should be selected and shown. Providing multiple alternatives can highlight the generative variability to the user.

If multiple outputs are available, the system should allow users to organize and label them, according to Weisz et al. [15]. The differences between the alternative outputs should also be highlighted to the user, the authors suggest. Furthermore, they encourage that a user's input history and corresponding outputs be saved and made accessible to the user, enabling iterative refinement and transparency in the generation process. Together, these design considerations aim to help users better understand, manage, and make effective use of generative variability.

Another principle established by Weisz et al. is that the designers should consider how to convey information about the generative AI system to the user, so that the user may form an appropriate mental model of the system [15]. A suggested strategy to achieve this is to inform the user about the stochastic nature of generative AI, meaning that a generative AI system may produce different outputs when given the same input. This could, for example, be shown by providing multiple output alternatives to the user.

To further increase the users' understanding of the system, Weisz et al. advocate for providing the users with tutorials and explanations of the system. Understanding the users' perception of the system and its functionality can help inform future design decisions [15]. However, Weisz et al. emphasize that the goal is not a completely accurate mental model of the generative AI system, but rather a useful and effective one.

Miller [19] similarly argues that explanations of an AI system are beneficial when forming a mental model of the system. Miller further suggests that explanations that align with human reasoning may be more useful than technically exhaustive explanations.

The AI should, in turn, have a practical understanding of the user, according to Weisz et al. [15]. The authors claim that designers therefore need to understand the system users and their values, to convey this to the AI. They further claim that teaching the AI about the user can allow for the generative AI system to better fulfill the users' expectations and wishes.

3.2.4 Appropriate trust and reliance

Weisz et al. [15] conclude that system designers should ensure the users' trust in the generative AI system is proportional to its trustworthiness, i. e., how accurate and correct its output is. This is corroborated by Lee and See [20], who establish the principle to "design for appropriate trust, not greater trust." To achieve this, Weisz et al. declare that users should be taught to carefully evaluate system output to determine if it is acceptable. However, designers can also help users form an accurate perception of the system's trustworthiness by explicitly communicating its reliability and limitations.

Another strategy given by Weisz et al. [15] is to show the origins of the generated output; for example, where the facts are fetched from and what materials the AI has been trained on. A previously mentioned strategy, showing users multiple output options to choose from, can also have an effect in this aspect; making users review different versions could make them reflect on its correctness and accuracy.

Similar to Zamfirescu-Pereira et al. [17], Weisz et al. [15] also encourage the designers to give the users a truthful depiction of the AI by not giving it too human-like characteristics, like using avatars and making it refer to itself as a person.

Lin et al. [16] also underline truthfulness as an important aspect, saying that designers must make clear what the system can do, and how well the system can do it in an objective

manner. This sentiment is echoed by Amershi et al. [21]. Lin et al. also state that users must be informed of the system's reliability and imperfections. Other recommended strategies to aid the user in trust calibration include providing verification and references for outputs, as well as comparing multiple outputs.

3.2.5 Designing for quality and responsible use

Designing generative AI systems responsibly involves anticipating potential harms of the system, using a human-centered approach, and managing unexpected system behaviors. To increase the chances of creating a useful system, Weisz et al. [15] suggest that stakeholders should be included in the design process. However, different stakeholders may have diverging values, and these discrepancies must be managed during the design process. One approach for managing such tensions is value-sensitive design, formalized by Friedman et al. [22], which emphasizes systematically incorporating human values into technological design.

Amershi et al. [21] highlight the need to mitigate social biases in AI output, as such biases can lead to unfair or discriminatory outcomes. The authors recommend identifying potential sources of bias in both training data and system design, and implementing strategies to monitor and reduce their impact.

Responsible design also requires considering how system functionality may allow or restrict unexpected behaviors. Emergent behaviors are not inherently negative or positive, and designers are encouraged to consider the consequences of allowing or restricting such behaviors by Weisz et al. [15]. Preventing harm caused by the system is another strategy emphasized, for example, by testing the system and creating mechanisms for handling harmful situations.

Another important aspect of responsible design is acknowledging that generative AI systems are inherently imperfect. Weisz et al. [15] encourage preparing users for the possibility that AI-generated output may not fully meet their goals or standards. This is important for managing expectations and preventing inappropriate levels of trust in the system. Weisz et al. recommend that the risk of inaccuracies be made explicit to the user, for example, through warnings or uncertainty indicators.

When possible, systems may also provide indicators of output quality, such as information about how closely generated content adheres to source material, say Weisz et al. [15]. However, such indicators should be presented carefully to avoid implying that correctness is guaranteed. To recover from potential accuracy issues, the system should provide tools such as editing, regeneration, or alternative outputs. It is also suggested that the users be able to provide feedback or criticism to the AI, for example, by indicating that an output is incorrect.

3.2.6 Integration into user workflows

In order for a tool to be useful, it must be easily incorporated into the existing workflow. If interacting with a system requires significant changes in habits and work practices, the required effort may discourage users from adopting the tool. Amershi et al. [21] therefore recommend making the invocation itself efficient, so as not to pose a barrier for the tool's usage. Similarly, Lin et al. [16] recommend integrating the tool invocation into the workflow.

Efficient workflow integration involves minimizing friction when interacting with the system. This aligns with general usability principles proposed by Nielsen [18], who argues that a system should be efficient to use by enabling users to finish tasks quickly.

3.3 Frameworks for measuring trust and acceptance

This section reviews widely used frameworks for explaining and predicting the acceptance and use of information systems. It begins with the Technology Acceptance Model (TAM) and its extension, TAM2, then summarizes the Unified Theory of Acceptance and Use of

Technology (UTAUT) and its extension, UTAUT2. For each framework, principal constructs and common measurement approaches are briefly noted.

3.3.1 Technology Acceptance Model (TAM)

Modern frameworks for user acceptance of information technology are commonly based on the Technology Acceptance Model (TAM) introduced by Davis [23]. This model is used to predict users' acceptance of a product, based on the theory that perceived usefulness (PU) is the primary predictor of user acceptance. PU is defined as "the degree to which a person believes that using a particular system would enhance his or her job performance" [23]. Another significant variable is perceived ease of use (PEOU), which is defined as "the degree to which a person believes that using a particular system would be free of effort" [23]. Davis also introduced a set of measurement scales to determine the PU and PEOU of a product.

The same paper suggests a relationship between PEOU and PU, namely that PEOU is a direct determinant of PU, thus indirectly influencing acceptance. In turn, Davis suggests that external factors such as training and system characteristics influence PEOU and PU.

3.3.2 TAM2

To account for the external factors influencing PU, TAM was extended by Venkatesh and Davis [24]. In their study, the authors presented a new framework called TAM2, which identified two main categories affecting user acceptance: social influence and cognitive instrumental processes. Social influence consists of three factors. The first is subjective norm, defined as "a person's perception that most people who are important to him think he should or should not perform the behavior in question" [24]. The second factor is voluntariness, described as "the extent to which potential adopters perceive the adoption decision to be non-mandatory" [24]. The third and final factor, image, is defined as "the degree to which use of an innovation is perceived to enhance one's status in one's social system" [24].

Cognitive instrumental processes are a category of PU determinants which is comprised of job relevance, output quality, result demonstrability, and PEOU. Job relevance concerns the extent to which a user perceives the system as applicable to their work, and output quality refers to how well the system performs its intended tasks. Result demonstrability indicates whether the benefits of using the system are visible or tangible. In TAM2, all aspects of the framework are measured through questionnaire items, as described in the original study [24].

3.3.3 The Unified Theory of Acceptance and Use of Technology (UTAUT)

The Unified Theory of Acceptance and Use of Technology (UTAUT), introduced by Venkatesh et al. [25], is another framework for predicting technology acceptance. UTAUT integrates elements from TAM, TAM2, and multiple other models, with the purpose of introducing even stronger prediction. UTAUT identified four core determinants of behavioral intention and use: the performance expectancy of the product, effort expectancy for the user, social influence of people surrounding the user, and facilitating conditions in the existing infrastructure.

The model was developed primarily for organizational settings, where new technologies are introduced to employees, and it also includes moderators such as age, gender, experience, and voluntariness of use.

3.3.4 UTAUT2

UTAUT was extended by Venkatesh et al. [26] by adding the constructs hedonic motivation, price value, and habit. These additions make UTAUT2 better suited to a consumer context rather than technologies introduced in the workplace, which was the focus of the original UTAUT framework. The voluntariness construct from UTAUT was also dropped due to this context change, as consumer behavior is typically voluntary.

Similar to the original UTAUT model, UTAUT2 aims to explain users' behavioral intention to use a system and their actual usage behavior.

3.3.5 Trust in Automation Scale (TIAS)

The Trust in Automation Scale (TIAS), introduced by Jian et al. [27], is a psychometric instrument for measuring trust in automated technology. In their study, Jian et al. compiled trust- and distrust-related words and evaluated them across three experiments. Using factor-analytic methods, they proposed a 12-item Likert-type measurement scale. Example items are "The system is reliable" and "I am wary of the system".

McGrath et al. [28] validated the TIA scale in AI contexts. The researchers also presented a shortened version of the TIA scale, the Short Trust in Automation Scale (S-TIAS), comprising three items from the original instrument. Moreover, S-TIAS was validated by McGrath et al. as a measuring tool of trust in AI.

3.4 Retrieval-augmented generation (RAG)

When generating content, large language models rely on knowledge acquired during training to produce responses: the so-called parametric memory. However, in domain-specific settings, it is often necessary to constrain the generation process so that outputs are based on relevant contextual information rather than the full breadth of the model's training data. One technique for achieving this is retrieval-augmented generation (RAG), which combines parametric and non-parametric memory [29]. This means that, in addition to knowledge learned in training, RAG systems can also use facts from external documents. By grounding generated outputs in retrieved documents, RAG has been shown to produce more factually correct and specific content compared to systems only using parametric memory [29]. RAG can also be applied to specific components of the generated output, for example, for image retrieval. It is important to note that due to the output being generated from retrieved data sources, any bias in the underlying data may propagate to the model's output. Consequently, the correctness, relevance, and quality of the underlying data are critical factors in RAG-based systems.

3.5 Related work

This section presents an overview of related work relevant to this thesis, including studies on educators' perceptions of generative AI as well as research on AI-assisted generation of learning materials.

3.5.1 Educators' perception of generative AI

Educators' perception of generative AI is central to the adoption of generative AI in an educational context, as attitudes towards usefulness and reliability may influence whether such tools are incorporated into educators' workflows. In a study from 2024, Ghimire et al. investigate university instructors' awareness of and attitudes toward generative AI and large language models in higher education [30]. The study is especially focused on the integration of AI in the classroom as a tool for learning, as seen from the educators' perspectives. These perspectives were derived using a Mixed Methods Evaluation, where the quantitative data were gathered via a survey and the qualitative data was gathered via follow-up semi-structured interviews. The study showed that the majority of the participating educators are aware of or have tried using AI tools, while about 40% of participants used AI tools regularly. The overall sentiment toward generative AI was cautiously positive among the participants, although some main concerns were identified among the educators, such as the risk of incorrect or fabricated results.

While Ghimire et al. [30] examine general perceptions of generative AI, AI-generated learning materials are not directly evaluated in their study, nor is there a focus on specific architectures or domains. In contrast to this prior work, the present thesis focuses on educators' perceptions of AI-generated lesson materials within a specific medical domain and evaluates their usefulness in relation to identified quality criteria.

Taheri et al. [31] conducted a meta-analysis examining factors influencing educators' willingness to adopt AI tools in the classroom. Based on 45 peer-reviewed studies, they found that these factors can be grouped into four categories: individual, infrastructure, impacts, and tools. The individual factors include demographics and AI literacy, while infrastructure encompasses institutional support as well as social environment and media. The category called impacts includes concerns about job displacement and over-reliance on AI. Within the tools category, PU and PEOU were identified as influential factors, along with the compatibility of AI products with users' existing routines and habits. Another critical factor for generative AI in education is the trustworthiness of the product. The users' perceptions of privacy and data safety were identified as important components of the trustworthiness of a product, as were concerns about ethics and AI bias.

TAM and UTAUT were the most commonly used evaluation frameworks in studies reviewed by Taheri et al. [31], which motivates the methodological approach of the present thesis. The emphasis placed on tool criteria such as PU and PEOU by Taheri et al. further reinforces the evaluation focus of this present thesis.

Al-Abdullatif [32] gathered the opinions on generative AI of over 230 university instructors via a structured questionnaire. The questionnaire was based on the TAM framework and was extended to include TPACK, a framework describing the types of knowledge teachers need to effectively integrate technology into their teaching [33]. In addition, Al-Abdullatif included AI literacy — defined as the educators' understanding of generative AI — into their framework. Lastly, Al-Abdullatif included perceived trust in the framework. In this context, trust encompasses aspects such as reliability, security, accuracy, and ethics.

The study showed a strong relationship between educators' PEOU of generative AI and their acceptance [32]. However, contrary to standard TAM predictions, no direct effect of PU on adoption was observed. It was shown that TPACK and perceived trust did affect the AI literacy of the instructors, but AI literacy didn't, in turn, have any direct effect on PU or PEOU. Moreover, both trust and PEOU were discovered to impact PU significantly. Lastly, it was shown that TPACK had a significant influence on PEOU.

These findings differ from the patterns suggested by the original TAM framework, indicating that TAM may not be fully applicable in this context [32]. Consequently, more context-specific validations of TAM and its extensions are warranted. This study is relevant to the present thesis because it highlights potential gaps in the TAM framework in the context of generative AI and suggests a way of extending TAM with other possible predictors.

Another study examining teachers' acceptance of generative AI in education was conducted by Hazzan-Bishara et al. [34]. In this study, 400 educators responded to a survey based on TAM extended with two additional categories of potentially predictive factors. One category consists of external factors: exposure to AI information, information credibility, and institutional support. The other category consists of internal factors, namely intrinsic motivation and self-efficacy. The results showed that exposure to AI information influenced perceived information credibility, which in turn influenced perceived usefulness. Additionally, institutional support was found to affect intention to use AI, as well as self-efficacy and intrinsic motivation. Furthermore, teachers' intention to use AI was associated with PU and PEOU. Other relationships were identified between intrinsic motivation and both self-efficacy and PU, and between self-efficacy and PEOU. The study by Hazzan-Bishara et al. is relevant to the present thesis because it applies an extended TAM framework to a similar but less specialized domain: generative AI in education.

3.5.2 Generating learning content with AI

The following studies examine different approaches to AI-assisted content generation, their benefits, and their limitations. Together, they provide context for the challenges and opportunities relevant to this thesis.

The automatic generation of programming exercises using LLMs was first examined in depth by Sarsa et al. [35]. Their work represents some of the earliest efforts to evaluate AI-generated learning content. This study investigated the feasibility of generating programming exercises and code explanations using the large language model (LLM) OpenAI Codex. The motivation behind the study was the intense workload put on educators in creating learning materials. By using prompts, the LLM could be used to generate learning material, which was then evaluated by the researchers themselves. The focus of this evaluation was the exercises' novelty, sensibility, and whether it was readily applicable or if they had to be manipulated by a human before use. In the first two regards, the majority of the generated exercises performed well. However, they would often require human intervention before they were ready to be used in teaching, often due to errors.

In a later study, Borys et al. [36] examined whether the generation of exercise content for mathematics and programming can be automated. In this study, an online learning platform called Compass was provided to a set of students who could use the platform freely. The platform incorporated human-made exercises, as well as exercises made with the help of an LLM. The AI-generated tasks were then evaluated by an expert and graded as one of the following three alternatives: *incorrect*, *partially correct*, or *correct*. A clear majority of the AI-generated exercises were deemed *correct*, while at most 1.6% were *incorrect*. The results showed that a larger percentage of the students passed their midterm exams, compared to the last year when Compass was not used. While this suggests that the platform—and potentially the AI-generated exercises—had a positive impact on learning outcomes, the study does not isolate the effects of AI-generated content from other platform features.

In a related study, Logacheva et al. [37] investigated the use of GPT-4 to generate exercises tailored to individual students' interests. The motivation for this study was that while student engagement can be enhanced by personalizing exercises, this action can be time-consuming and burdensome for educators. The generated exercises in this paper were evaluated by the researchers in terms of correctness, clarity, and appropriateness. Additionally, the participating students also gave feedback on how they perceived the usefulness, enjoyment, engagement, and how well the exercises matched the selected interests of the students. The results of this study were positive: the overall response from the students was positive, which was in accordance with the researchers' evaluation of the exercises. However, one issue encountered was that the generated material sometimes came out faulty, again indicating the need for human oversight.

These studies are examples of how LLMs have been used successfully in different contexts to generate learning materials. Despite the often positive results, these studies also highlight limitations such as errors in the generated material and the continued need for human intervention to ensure quality. While prior research has largely focused on technical feasibility and student-level outcomes, less attention has been paid to how educators view the integration of AI-generated exercises and lessons into their teaching. This thesis addresses this gap by examining educators' views on AI-generated lesson materials within a medical domain and by evaluating their usefulness in relation to explicitly defined quality criteria.

4 Method

This project was divided into three main phases: the pre-study phase, the implementation phase, and the evaluation phase. The pre-study phase consisted of a literature search to identify related work and evaluation frameworks, as well as an initial round of user interviews to uncover user needs. The implementation phase followed, in which prototype features were designed and implemented based on insights from the pre-study. Lastly, the evaluation phase entailed a product evaluation with a larger group of participants. These phases are described in Sections 4.1, 4.2, and 4.3, respectively.

4.1 Pre-study

The pre-study consisted of a literature review and an initial round of qualitative interviews. The purpose of the interviews was to guide the implementation by uncovering development areas identified by the users. This chapter describes the procedures of these two initial phases, as well as the analysis of the interview data.

4.1.1 Literature search

One important goal of the pre-study was to select an appropriate evaluation framework to use in the evaluation phase of the project. Thus, one focus area of the literature review was finding well-established evaluation frameworks for information technology systems. The most relevant of these frameworks were identified and described in Section 3.3.

The primary criteria for selecting a framework were that it should be validated in an information technology context and suitable for evaluating early-stage prototypes. For example, TAM2 introduces motivators such as social influence, which are more relevant in contexts where systems are already integrated into regular use. As the prototype evaluated in this study is not yet deployed in a real-world setting, these factors were considered less relevant. Furthermore, this study focuses on perceived usefulness and perceived ease of use, which are central constructs of TAM. A simpler structure was also deemed more suitable, given the relatively small number of participants in the final evaluation (N=15; see Section 4.3), as more variables generally require a bigger dataset to produce statistically meaningful data. For these reasons, the Technology Acceptance Model (TAM) was selected as the evaluation framework for this study.

Trust was also deemed an interesting factor for generative AI systems in the medical education domain. Therefore, a framework for trust evaluation was also needed. The criteria for selecting this framework were that it should be validated in the context of automated or AI-based systems and have a small number of questionnaire items. The latter criterion was motivated by the limited testing time available for participants. The criterion was met by S-TIAS (see Section 3.3.5).

The search for frameworks was primarily conducted through Google Scholar. Examples of search terms include "educator AI adoption", "educator generative AI attitudes", "trust in artificial intelligence scale", and "measure trust in artificial intelligence".

The design principles described in Section 3.2 were extracted from peer-reviewed literature. This search, too, was primarily conducted in Google Scholar, using search terms such as "AI design principles" and "usability heuristics".

Related works were discovered in the same manner as described above: by searching Google Scholar for terms such as "educator AI adoption" and "educator generative AI attitudes". Some evaluation frameworks were also encountered in the related work search, as studies examining educators' acceptance of AI systems naturally reference the frameworks used to measure it.

4.1.2 Participants and recruitment

Participants were recruited from existing Sectra Education Portal users through internal contacts at Sectra, including customer success managers and a product manager. The internal contacts identified potential participants based on prior experience creating lessons manually and their regular use of lessons in educational contexts. The goal was to as accurately as possible represent the educators using Sectra Education Portal. Since the knowledge base underlying the system primarily contained radiology and histology images, recruitment was intentionally focused on educators within these domains. This was done to ensure that participants could evaluate the tool under conditions where the generated content was likely to be more accurate and relevant, thereby enabling a fairer assessment of system performance.

Six participants took part in the pre-study interviews: four in individual interviews and two in a paired interview. Two individual interviews were conducted via online video call, while the remaining interviews were held in person. Five out of the interviewed persons were educators in higher education who regularly use lessons in their teaching, with teaching areas including occupational therapy, radiography, and histopathology, among others. The sixth person regularly created lessons manually for educators but did not teach themselves. Their experience with creating lessons in Sectra Education Portal ranged from none to extensive.

4.1.3 Interviews

The purpose of the interviews was to explore users' perceptions of the existing lesson generator prototype. To achieve this, participants in the individual interviews were given a brief introduction to the lesson generator prototype before being asked to generate a lesson with it. They were instructed to create a lesson they might realistically use in their profession, and to express their thought process aloud as they proceeded. After using the prototype, participants were asked follow-up questions about usability, perceived quality, and trust. Example questions included how participants experienced the lesson generation process, whether they would use the generated lesson as is, and what they would like to change about the prototype at this stage. See Appendix A for the full interview guide. The interviews were semi-structured and exploratory, allowing follow-up questions based on participants' responses. The prototype was still available to the participants for testing or demonstrating as they were answering the questions.

The in-person interviews were audio-recorded and transcribed after the sessions. For the online interviews, transcripts were generated live using the built-in transcription functional-

ity in Microsoft Teams and subsequently reviewed and edited for accuracy. Field notes were recorded for all interviews. All data collection was conducted with verbal consent from the participants.

Due to time restrictions, the paired interview had a slightly different structure. In this session, the researcher demonstrated the prototype by generating a lesson. The participants were then asked questions in the same manner as described above. The meeting was documented in notes with the participants' verbal consent, but was not audio recorded.

4.1.4 Data analysis

The analysis followed the six phases of thematic analysis, as described by Braun and Clarke [38]. The methodology is described below.

In the first phase, the researcher familiarized themselves with the data. During this phase, the audio recordings were transcribed, and the participants were anonymized. Participants are henceforth referred to as P1, P2, P3, and so on. The transcripts were read repeatedly, and initial ideas were noted.

Phase two entailed generating initial codes, which are labels capturing important features of the data relevant to the research question. In the present project, the coding was conducted manually, but it can also be done with the help of different software tools. Although this phase appears early in the sequence, the coding may be revisited at later stages of the thematic analysis as well. In the case of this thesis, the labels were refined at several points in time throughout all subsequent phases.

The third phase required grouping the previously defined codes into broader themes by analyzing the relationships between the existing codes. The themes were then reviewed in phase four. The review was conducted at the code level first, i.e., it was analyzed whether the codes fit together. At this stage, some codes were rewritten and regrouped, and some themes were redefined. This was followed by a review of the themes in relation to each other, thus analyzing the categorization of the entire data set. This was done iteratively, meaning it was conducted in several different rounds.

After the themes were mapped and refined, they were thoroughly defined and analyzed in phase five. In this step, the scope and definition of each theme were determined. Finally, in phase six, a report of the analysis was written, which is located in Section 5.1.

4.2 Implementation

This section explains the implementation process, starting with the development process: what languages and libraries were used and how features were selected for implementation. Next, the methodological design decisions of each implemented feature are explained.

4.2.1 Development process

The extended prototype was developed using React with Typescript together with C#. For AI integration, Azure AI Foundry was used. GitHub Copilot and Claude were used as coding tools to discuss possible solutions and to debug. As the implementation was a prototype, the scope was limited to features that could be realistically developed and evaluated within the time frame of this thesis.

Features were implemented one at a time, in order of priority. Functionality related to the most popular themes from the initial interviews was prioritised, followed by less frequently mentioned themes. Within each theme, the frequency of individual issues raised by participants was also taken into account. Each feature was developed iteratively. Medical Education team members were consulted initially when designing the planned feature, resulting in adjustments to the intended design before implementation began. The feature was then developed, after which team members, including the external supervisor of this thesis,

were consulted again, and further refinements were made based on their feedback. These steps were repeated until the team deemed the feature sufficient.

4.2.2 Feature design and selection

The feature selection was driven by the results from the pre-study interview round, as well as the design principles described in Section 3.2. An overview of the pre-study themes and the feature implementations they led to can be seen in Table 4.1.

Pre-study theme	Feature(s)
Control and Revision	AI chat, Lesson plan, Updated prompt form
Usability	Updated prompt form
Trust and Transparency	AI Provenance Banner
Impact on Existing Workflow, Pedagogical Quality	Batch-based generation

Table 4.1: Pre-study themes and their corresponding implemented features.

The pre-study theme Interaction and Feature Expansion was not addressed in this implementation. This is because the issues raised under this theme fell into three categories which were outside the scope of this thesis: image selection problems rooted in the image metadata in the underlying knowledge base (which would require medical expertise to address), model behavior issues, and requests for product-level functionality. Below, the methodological design decisions of each feature are described.

AI Chat: Users requested being able to iteratively revise the lesson using AI, as stated in Section 5.1.1. Lesson revision could be supported, for example, by providing a field for re-prompting or a feedback form. However, one user explicitly mentioned wanting conversational AI, as they were already familiar with chat-based AI tools. Additionally, a chat provided versatility in being able to handle a multitude of different revision requests. Therefore, the AI Chat was implemented. It was placed in the bottom right corner to avoid obstructing the view of any lesson contents.

The AI Chat was implemented first and received the most development effort, reflecting that users explicitly and repeatedly requested AI-assisted revision during the pre-study interviews. The AI Chat was also designed to align the tool with the principles in Section 3.2.1: allowing the user to co-edit lessons with the AI and to refine individual elements rather than regenerate the entire output. To avoid making the agent appear too human-like, as discouraged in Sections 3.2.4 and 3.2.1, it was not given a name or persona but was called "AI Chat". The chat also preserves past requests and enables the user to revert or reapply the changes triggered by that request. These features implement principles mentioned in Section 3.2.3, specifically preserving the user's input history and the system's corresponding output.

Updated prompt form: As shown in Section 5.1.1, users frequently requested a more guided generation process, to get a better understanding of what to include in the prompt. Users also requested example prompts. Both of these needs align with usability design principles described in Section 3.2.2. Prompt examples were therefore added to the form. To make the prompt form more guided, more input fields were added, based on specific requests from the interview participants. As some interview participants also found the wording of the form ambiguous, several fields were reworded with the help of Sectra Medical Education team members, including the external supervisor of this thesis. The purpose was to choose field titles that were clearer and easier to understand. As users' proficiency levels when using AI tools may differ, freehand prompting was added as an option; for example, experienced users may already know how to prompt effectively and might prefer writing their own prompt.

Lesson plan: As shown in Section 5.1.1, this feature was directly requested by users. It was also a way of achieving collaboration between the user and the AI in the revision process, aligning with design principles in Section 3.2.1. Another reason to implement the lesson plan was to enable user feedback on the lesson earlier in the process, before the full lesson is generated. This was preferable because regenerating a plan is faster than regenerating a full lesson, and less data is needed, reducing latency. This is also consistent with design principles in Section 3.2.6 on efficient invocation. Through this feature, users are encouraged to give feedback to the AI system, which is advocated in Section 3.2.3 and Section 3.2.5. Section 3.2.5 also says that there should be tools for generation and alternative output, which this feature provides.

AI Provenance Banner: In the initial interviews, users mentioned wanting to know why a certain image was selected over another. Showing the origins of the output is also a design principle mentioned in Section 3.2.4 to encourage appropriate trust in the system. Therefore, an AI Provenance Banner was added, which displayed the search terms used by the AI agent, as well as why the image was selected. To keep the disclaimer linked to the slide's image, while not blocking the view of it, the banner was placed in the slide, between the title and the description.

Batch-based Lesson Generation: In the initial interview round, system latency was noted by several users, see Section 5.1.6. To reduce the time needed to generate a lesson, batch-based lesson generation was introduced. Another reason for this change was interview participants finding the output had too little information, see Section 5.1.4. By generating batches of a few slides each independently, the agent could produce more detailed slide contents without needing to fit the entire lesson into a single response, reducing the risk of truncated or overly compressed output due to token limitations.

4.3 Evaluation

The evaluation phase assessed how medical educators perceived the usefulness, ease of use, and trustworthiness of the two prototypes. Participants completed a set of tasks in both prototypes, after which they filled in a questionnaire based on the TAM and S-TIAS frameworks. Finally, a qualitative interview was conducted with each participant to gather more in-depth insight into their perceptions and experiences of the two prototypes. The evaluation procedure is described in detail in the following sections.

4.3.1 Participants and recruitment

Participants for the final product evaluation were recruited in the same manner as for the pre-study interviews: with the help of internal contacts at Sectra. Potential participants were identified with the intent to be as representative as possible of Sectra Education Portal users who create or use lessons. Their experience with the platform and the lesson structure was considered, as were their areas of expertise and the level at which they teach. The available materials in the knowledge base were primarily radiology images (such as X-ray and CT scans) and histology images (microscopic images of tissue). To enable a fairer assessment of system performance, recruitment was focused on educators in these areas.

In total, 15 users participated in the final evaluation. Participant experience ranged from no prior lesson creation to extensive experience; the exact distribution can be seen in Figure 4.1. While most users were located in Sweden, participants were also recruited from South Africa, New Zealand, the UK, Denmark, and Norway. Three of the participants were Sectra employees who do not currently teach. However, one of the Sectra employees creates lessons for other educators, while the others are in direct contact with educators and have a unique

perspective on their needs and opinions. All other participants were educators, with teaching levels ranging from undergraduate to specialist training to continuing professional development. The teaching areas represented were mainly radiology, but also included histology, anatomy, and pathology, among others. The participants were assigned IDs P1-P15, and are henceforth referred to as such.

Out of six participants in the initial interviews, five participated in the final evaluation as well. They were given the same IDs upon analysis: P1, P2, P4, P5, and P6. All participants were informed that their participation was voluntary, that they may withdraw their participation at any point without a specified reason, and that any data associated with them will be deleted upon their request. No compensation was provided.

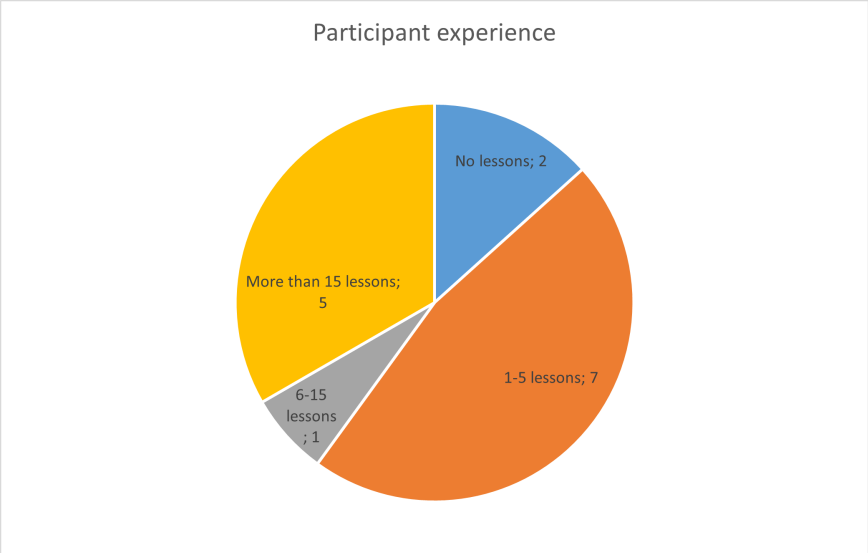


Figure 4.1: Participants' experience distribution, based on how many lessons they had previously created in Sectra Education Portal.

4.3.2 Design and setup

The evaluation compared an original prototype with an extended prototype to assess differences in perceived ease of use (PEOU), perceived usefulness (PU), and trust. To achieve this, the evaluation was composed of two activities: prototype testing and interviews. Participants received standardized written instructions. To control for order effects, eight participants used Prototype A (extended) followed by Prototype B (original), and seven followed the opposite order. Prototypes were labeled A/B to avoid bias, and participants were not told which version was original or extended. To control for confounding factors, both prototypes used the same language model (GPT-5 mini) and knowledge base configuration; only feature availability differed. Sessions were unmoderated and conducted remotely using test accounts in Sectra Education Portal on participants' own devices and browsers.

4.3.3 Procedure and tasks

For each prototype, participants completed the following tasks:

1. **Generate a lesson**: Imagine you are preparing a lesson for use in your role. Choose a topic you are familiar with and generate a lesson using either the guided form or the freehand prompt. Take a moment to review the quality and accuracy of the content before moving on.

2. **Edit and refine the lesson**: Modify and improve the lesson. Examples of changes you can try: Edit or rewrite text, adjust or replace questions, change the structure, or adapt to a different difficulty level

3. **Try a more targeted request**: Generate another lesson on a different topic. If the first lesson was too general or too short, be more specific or make this one longer. The goal is to explore the tool further and apply what you learned from your first lesson.

For Prototype A (extended), participants were instructed to use the AI Chat at least once to try out the extended functionality.

4.3.4 Measures

After finishing the tasks for each prototype, the users were asked to fill in a Likert-scale questionnaire consisting of three parts: six TAM-style items about perceived usefulness, six TAM-style items about perceived ease of use, the 3-item Short Trust in Automation Scale (S-TIAS) [28], and one additional trust-related item assessing the perceived need to verify AI-generated output. All Likert scales used seven-point response options. TAM items used "Extremely likely" (1) to "Extremely unlikely" (7) anchors, and trust items used "Strongly agree" (1) to "Strongly disagree" (7). Free-text fields captured additional comments; see the full evaluation form in Appendix B. Participants could complete tasks within a three-week window, but were instructed to submit the questionnaire immediately after trying each prototype, in order to capture their initial impressions.

4.3.5 Interviews

Lastly, the evaluation also entailed a qualitative interview held in the weeks after all previous parts were finished. The purpose of the interview was to acquire qualitative data on user opinions of the different prototypes, to provide context for the questionnaire scores. Three of the participants were interviewed in person, and all other participants were interviewed over a video call in Microsoft Teams. Interviews were recorded with the consent of each participant – in-person interviews were transcribed after the interview, while the video calls were transcribed live with Microsoft Teams' built-in transcription functionality.

Participants were first asked about their overall experience and how the two prototypes compared. This was followed by questions about perceived ease of use, such as "How did you find the process of generating a lesson with the two different prototypes?" and "What was confusing or frustrating about the prototypes?" The next segment of questions was about usefulness and included questions like "How would you feel about using either prototype in your day-to-day work?" Trust was explored via questions like "What made you trust or distrust the output?" Lastly, the participant was asked if they would like to share anything else before the interview finished. See Appendix C for the full interview guide.

4.3.6 Data analysis

The final construct scores were computed as the mean of the corresponding items per participant. Standard deviations were calculated across participants to find variability between user experiences, using the sample standard deviation. The interview transcripts were then reviewed manually, and statements related to usefulness, ease of use, and trust were identified and grouped. This qualitative data was used to better understand why participants evaluated the prototypes differently, and to provide illustrative examples and explanations for patterns observed in the questionnaire responses.

5 Results

This chapter presents the results of the three project phases. First, the themes identified in the pre-study interviews are described. Next, the extensions implemented in the prototype are detailed. Finally, the evaluation results are presented: first, the questionnaire scores, then the findings from the final interviews regarding perceived usefulness, perceived ease of use, and trust.

5.1 Pre-study

In the first phase of the project, a round of initial interviews was held, as described in Section 4.1.3. From the interview data collected, six main themes were identified: Control and Revision, Trust and Transparency, Usability, Pedagogical Quality, Interaction and Feature Expansion, and Impact on Existing Workflow. The distribution of mentions (N=140) was led by Control and Revision (33; 24%), followed by Trust and Transparency (27; 19%) and Usability (24; 17%); Pedagogical Quality (23; 16%), Interaction and Feature Expansion (17; 12%), and Impact on Existing Workflow (16; 11%) were also prominent. These statistics are illustrated in Figure 5.1, and the themes are described in the following subsections.

5.1.1 Control and revision

The most common theme in the interviews was Control and Revision, which is centered around the lesson generation process and the users' desire to influence it in order to refine the generated content. The composition of the theme is shown in Figure 5.2.

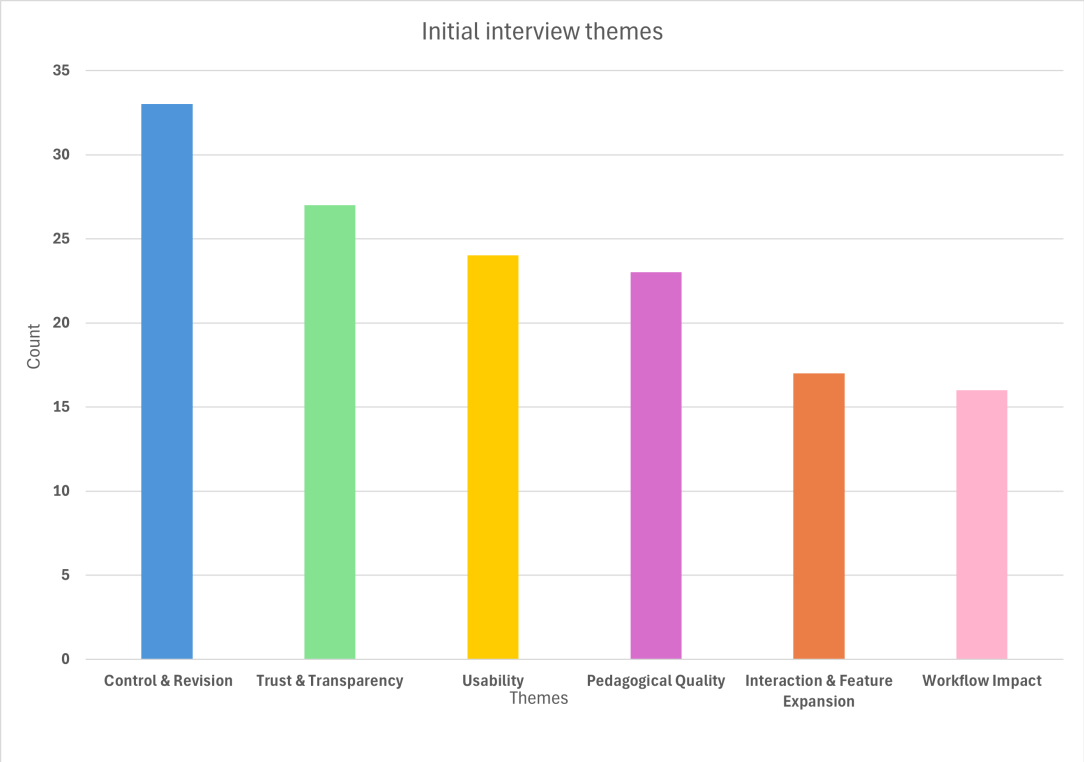


Figure 5.1: Bar chart of themes identified in the initial interview data. The figure shows the total number of mentions per theme.

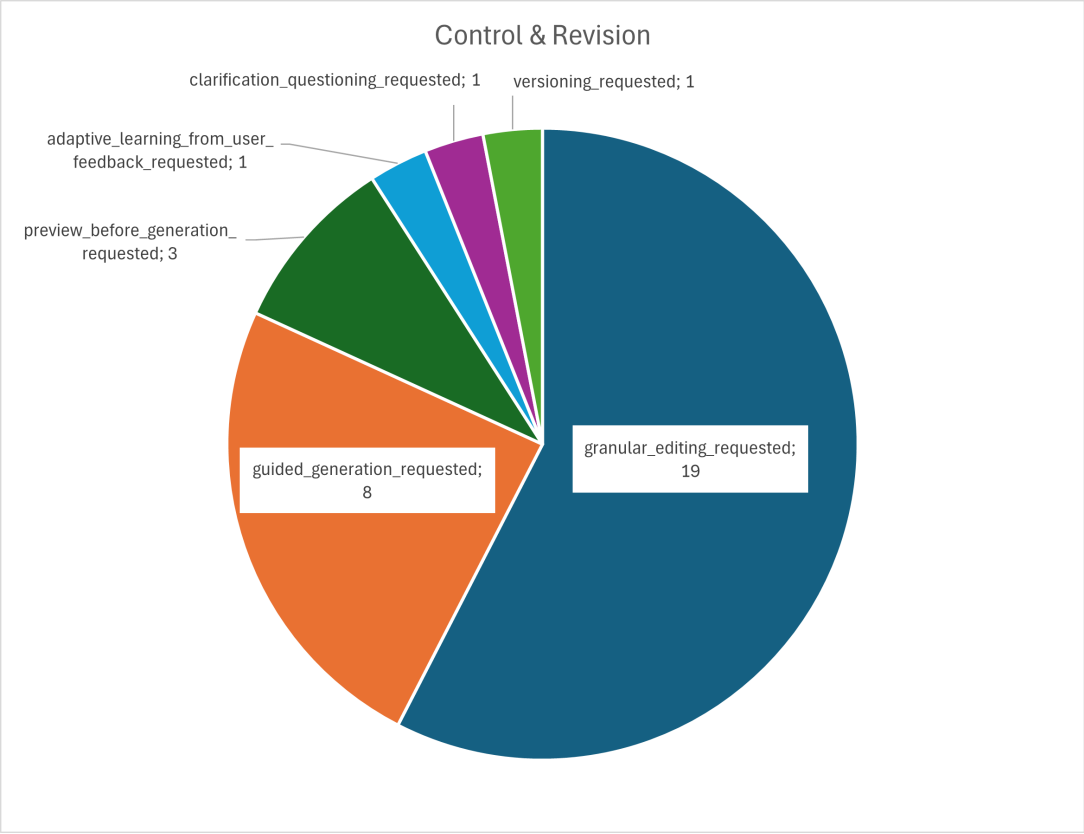


Figure 5.2: The tags composing the Control and Revision theme, and their mention count in the initial interview data.

Primarily, this theme encompasses the need for a more granular revision process. For the first prototype, revision was only possible manually. While most participants expressed that they would use manual revision, one of the most common requests was the ability to regenerate the lesson, or specific questions in it, by giving direct feedback to the agent. For example, P1 said the following: "I think that, in order for me to feel that this is really a new way of working or a new tool, I would probably like to be able to go back to the prompt and change it." P6 similarly stated: "I think that being able to give direct feedback would have been good. That I can say: no, question number one doesn't work, do it again."

Participants also mentioned the need for more guiding in the generation process, to improve the quality of the generated lesson and reduce the need for revision. This highlights a tension: users want more system guidance to avoid poor outputs, but still want to retain control over final content. P5 suggested "Instead of writing a prompt and hoping for the best, then maybe just put a filter in and say 'I'm only interested in these kind of images'." Example prompts were also requested by a participant.

Although some interview participants requested more guidance when prompting the lesson generator, several also stressed the need for the user to control the process. Suggestions were made to show previews of the generated material and selected images, in order to permit the user to accept or reject certain material.

P4 suggested giving the agent the ability to learn the most common reasons for user dissatisfaction. The same participant also wanted the agent to be able to ask clarifying questions if the prompt was too ambiguous or unclear. Versioning was also requested, so that users could go back to older variants of a revised lesson.

Taken together, these findings suggest that participants would value AI lesson generation more if it supported iterative, user-led refinement rather than one-shot generation.

5.1.2 Trust and transparency

The Trust and Transparency category was the second most common theme, and it encompasses concerns about how the system actually works and how trustworthy the generated material is. Figure 5.3 illustrates the theme's composition.

The main issue in this theme was regarding the system opacity; users were wondering how much impact their prompt actually had on the generated lesson, how the images were selected to be used in the lesson, and how the agent constructs different lesson types. This was, for example, expressed by P1 who said "I might say that I don't really know how important it was to write a prompt." P4 said the following about an image in a generated lesson: "This one is available as a CT as well, so it's interesting to know why [the agent] chooses a 3D model in this case." P4 also argued that users might trust the tool more if it was explicitly stated where the images came from.

Another major concern for the users was trust calibration. Multiple participants mentioned that they felt the need to review the relevancy and correctness of the generated material, rather than trusting the tool completely. P2 and P3 noted that their students might lack the expertise to critically review the lesson generator output. P6 believed that one's trust in AI can depend on what generation one belongs to, and that their students were more likely to trust the lesson generator's output than the participant was. However, several interview participants also said that they have trust in the underlying material, i.e. the medical images and their descriptions.

P4 expressed that the lesson generator would be more trustworthy if it had been validated by an expert, and if the underlying material had been reviewed by a board of experts. P1 also questioned how accurately the material was labeled in the knowledge base.

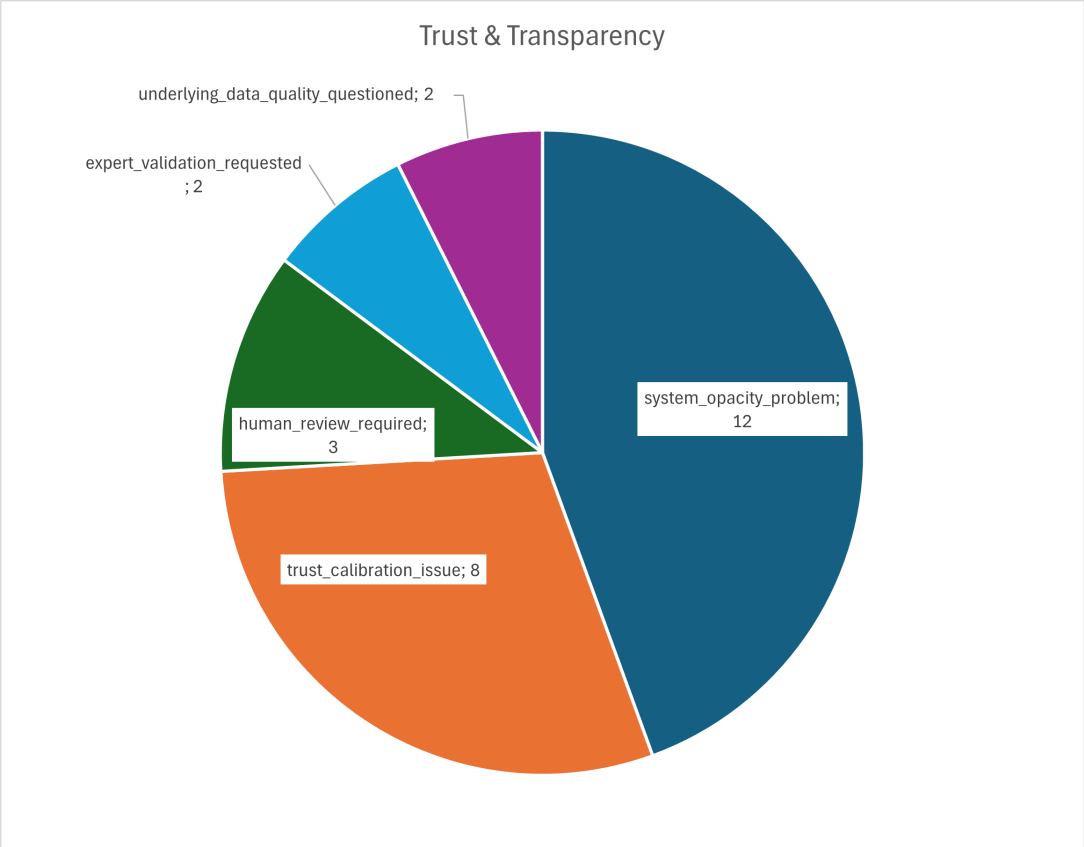


Figure 5.3: The tags composing the Trust and Transparency theme, and their mention count in the initial interview data.

5.1.3 Usability

The theme Usability was the third most mentioned, and it includes aspects concerning how easy the system is to use and learn. The theme's contents are shown in Figure 5.4.

P5 especially emphasized prompting as a learned competency rather than a self-explanatory input method, saying "You have to practice writing good prompts." However, several users mentioned that the tool was similar to chatbots that are already included in their existing workflows, and that this made the lesson generator easier to use.

Multiple participants brought up that there must be low friction when using the tool, meaning that it must be easy to use and it must produce high quality output. An example of this is the following quote from P6: "If you are going to use it, it must be easy to use." This was corroborated by P4 who said "My experience is that the time a teacher has to sit and play around is non-existent. It must work immediately in a live setting."

Some users found the terminology used in the generation process ambiguous. Concepts such as "learning objectives" and the names of different lesson types caused confusion for several of the participants. Some thought that the interaction was intuitive. P4 pointed out that a long lesson title is cut off in the user interface and can't be shown in full.

5.1.4 Pedagogical quality

This theme captures how well the generated lessons were perceived to support teaching goals and instructional practice. The data show that multiple participants thought that the structure of the generated lessons was generally good, see Figure 5.5. For example, P1 said the following about the order of images chosen by the agent: "Pedagogically, this is the best choice."

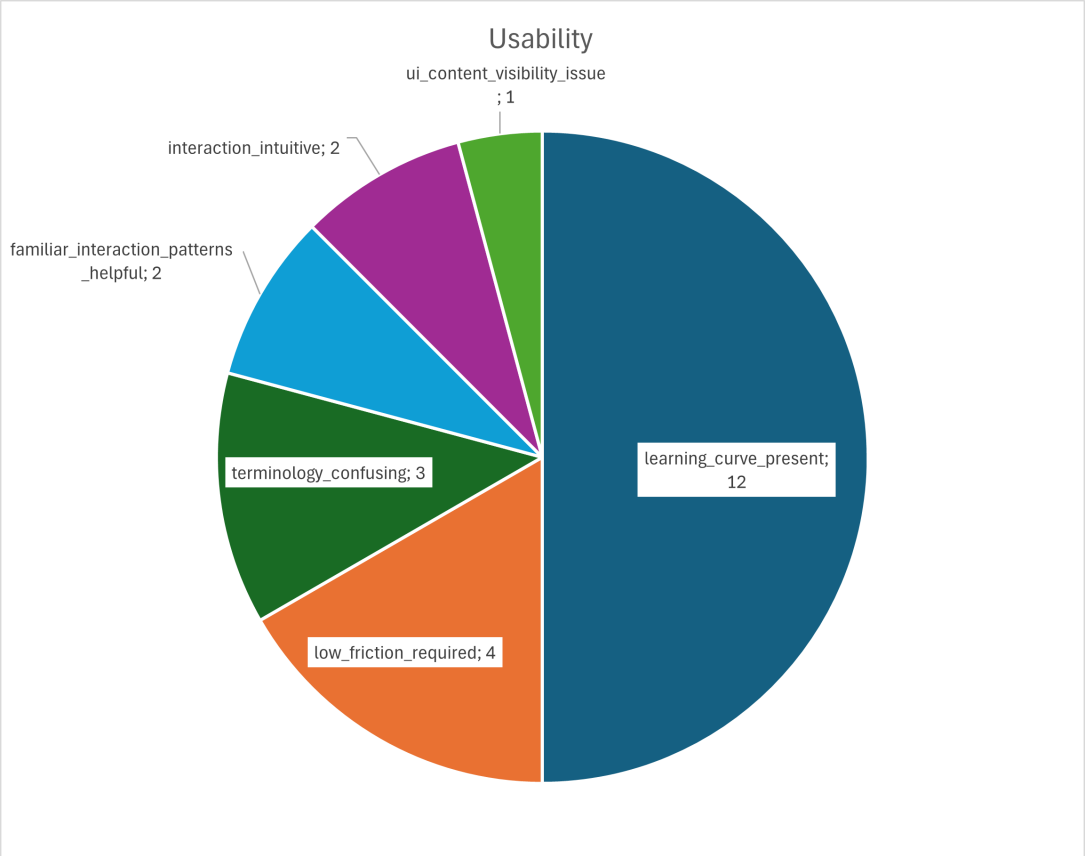


Figure 5.4: The tags composing the Usability theme, and their mention count in the initial interview data.

Similarly, P6 expressed that the generator selected the same images that the participant had used in their own teaching. P1, P2 and P3 stated that the generated lesson provided a useful lesson draft.

P6 noted that the content should be more adaptable to different working fields, as the generated lesson was not completely relevant to their specific field: "It could be good to adapt the level to the profession." P1 commented that their prompting strategy would differ across educational settings, suggesting a need to align the generation process more closely with educators' specific contexts.

The pedagogical context was also mentioned by several participants, who wanted more information in the lesson slides. For example, P4 indicated this in the following statement about a generated case study lesson: "Since it's a case study, it would be interesting if it said a little more in each slide, like what it is that you are expected to do, why it is important to look at these parts." P5 and P6 shared similar sentiments. Despite these omissions, participants still found the generated content relevant. P5 said about the questions: "They were very relevant and on topic."

A few content-quality issues were reported by P4 and P6. P4 generated a lesson in which an image with messy annotations was selected. There was also a lesson slide that used the term "pathology" incorrectly, as noted by P4. In P6's case, the lesson was generated in Norwegian, but it used an uncommon medical term instead of the standard Norwegian term. Participants also requested formatting improvements, such as headings and additional spacing.

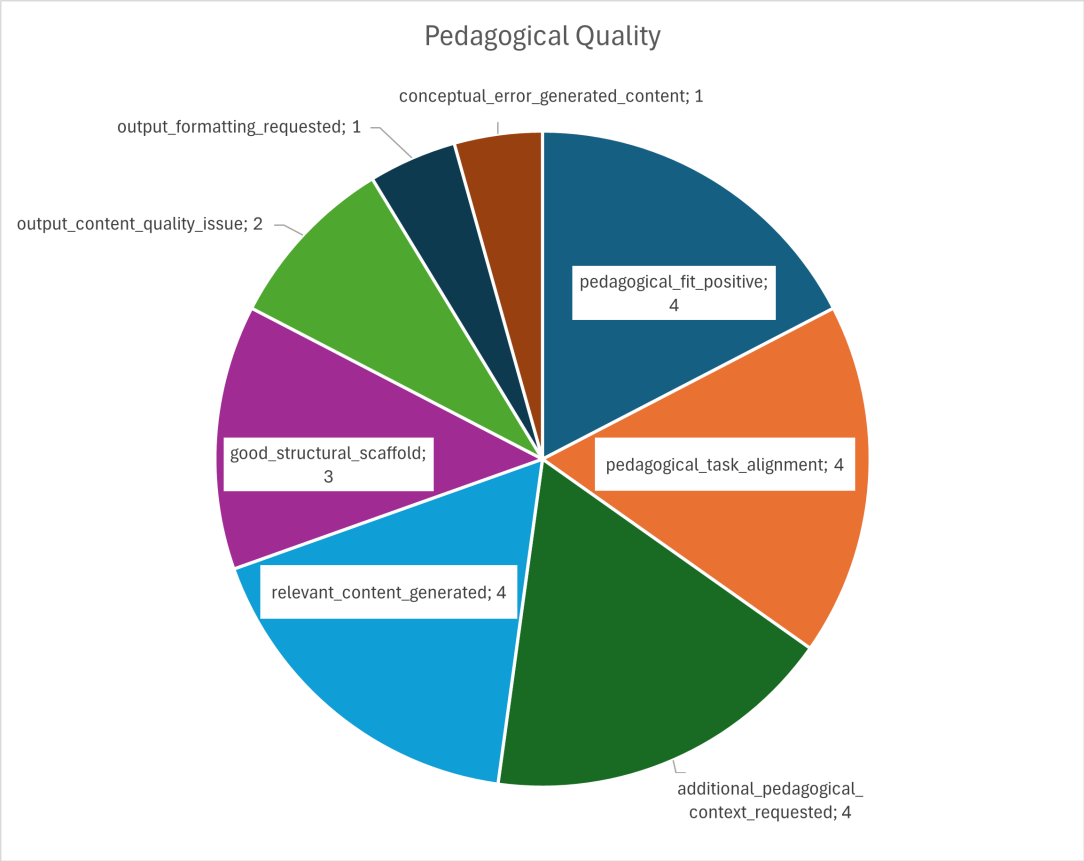


Figure 5.5: The tags composing the Pedagogical Quality theme, and their mention count in the initial interview data.

5.1.5 Interaction and feature expansion

This theme captures usability problems in interacting with the lesson generator as well as requests for additional functionality, see Figure 5.6.

There were sometimes problems with the image selection, such as atypical images being chosen, including cases where an image showed a relevant pathology alongside an unrelated finding. P1 commented: "Then, as a teacher, you might want to have a text explaining that, in this particular case, there is a clot in the right pulmonary artery, but that this is not the focus of this lesson slide."

Additional unexpected behavior was observed. For instance, the system selected only lower-extremity images even though this was not specified in the prompt. In some cases, descriptions referred to findings that were not visible unless the user adjusted the viewing settings. One participant asked for Latin terminology only, but the system misinterpreted this and generated the entire lesson in Latin.

Beyond these interaction issues, participants suggested several feature additions to Sectra Education Portal. P1 requested richer question types, such as enabling students to identify a body part by clicking on the image. P4 requested personalization features. These included aligning the generated material with the educator's personal style or the institution's typical lesson structure.

Other requests included the ability to export lessons, select multiple images per slide, and allow the agent to access files from the user's storage.

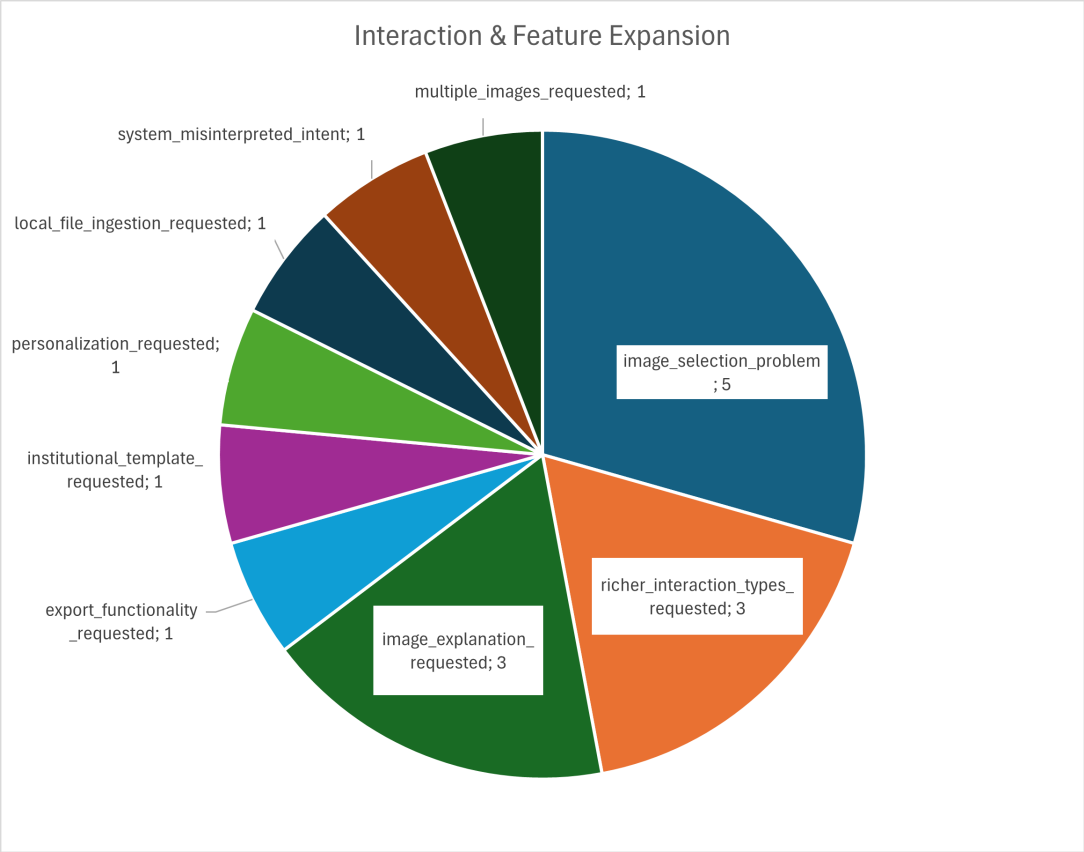


Figure 5.6: The tags composing the Interaction and Feature Expansion theme, and their mention count in the initial interview data.

5.1.6 Impact on existing workflow

This final theme focuses on how the tool fits into participants' existing workflows, see Figure 5.7 for the theme contents. P2, P3, and P4 reported that they usually revise existing material rather than create lessons from scratch. P4 suggested the tool might encourage more from-scratch creation: "It may be that now that there is the opportunity to start completely from scratch, you start doing that."

Several participants anticipated efficiency gains when creating new lessons. P6 said the following: "When using this, I can create 40 questions in half an hour instead of sitting for three hours and coming up with things. I think it's easier for me to create a lot of questions." P5 also commented on the possible efficiency gain: "The most useful thing is that you don't have to plow through the entire library to find an examination you can use. It can find some very useful examinations for you."

However, system latency could reduce these previously mentioned efficiency gains. This was noted by several of the participants. For instance, P5 said, "It can take some time for it to generate a lesson." Participants also raised reliability concerns, such as error messages during lesson generation.

P4 noted that suboptimal generation could lead to abandonment of unfinished lessons: "When I've tried different AI tools myself, I often end up with 25 half-finished things. And then you quit because it won't be good enough to use it. And then you don't have the energy to continue."

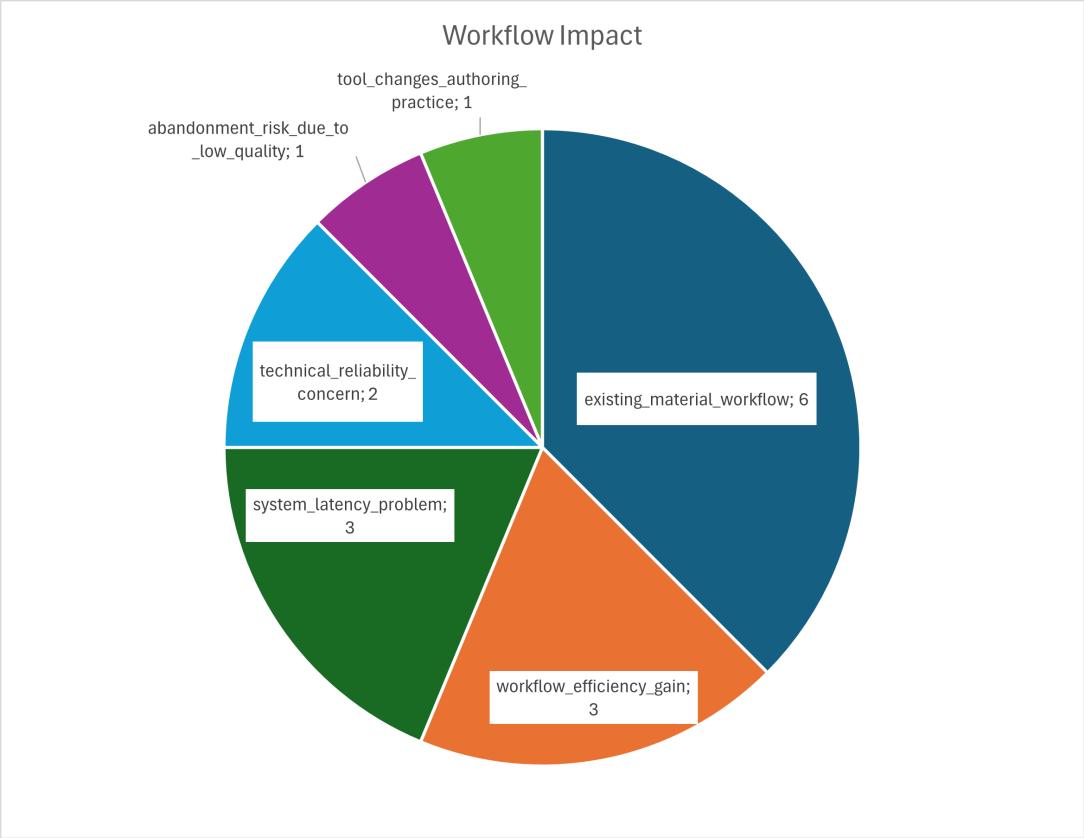


Figure 5.7: The tags composing the Impact on Existing Workflow theme, and their mention count in the initial interview data.

5.2 Implementation results

At the start of this project, an existing prototype for lesson generation already existed in Sectra Education Portal as described in Section 2.2.3. This prototype served as the foundation for the work carried out in this thesis. After the results of the pre-study interviews had been analyzed, the extensions of the prototype were designed and implemented as described in Section 4.2.

This section describes the features included in the extended prototype, i.e., what was added to the original lesson generator. Based on the initial interview round and established design principles, the following features were implemented: an AI Chat for lesson revision, an Updated Prompt Form, a Lesson Plan that is displayed before lesson generation, and an AI Provenance Banner for images selected by AI. The backend code for the lesson generation process was also changed. The purpose of changing the backend code was to increase the perceived usefulness by decreasing the time needed to generate a lesson and by enabling more detailed lessons. The AI Chat and the Lesson Plan were also implemented to extend the prototype's support for perceived usefulness by giving users more control over the output. Additionally, the AI Chat was intended to improve the ease of use of the prototype by reducing the effort required for lesson revision. The perceived ease of use was also targeted by the Updated Prompt Form, as it was intended to reduce the cognitive load of prompt construction. Lastly, the AI Provenance Banner was implemented to make AI decision-making more transparent in order to increase trust.

5.2.1 AI chat

Feature description: When editing a lesson, an AI chat is now available, where the user can request changes to the lesson or ask questions about the content. When the AI makes changes to a lesson, the affected slides can be individually accepted or rejected. Alternatively, all suggested AI changes can be accepted or discarded collectively via buttons in the chat. Accepted AI changes can also be reverted and reapplied via the chat. An example AI chat conversation can be seen in Figure 5.8. A full-screen view of suggested AI changes can be seen in Figure 5.9.

System details: When the user sends a message, a snapshot of the lesson's current state is saved to enable reverting. Then, a call is made to a commercial LLM (GPT-5 mini) to decide how to handle the message. The requested changes are then carried out by a revision agent using the same model, which has access to a knowledge base of Sectra's medical image metadata. The model classifies user requests according to the types below.

- BATCH_EDIT: The request affects many or all slides, so the lesson slides will be split into batches to be handled concurrently by the revision agent.
- SINGLE_EDIT: The request affects single slides or general lesson info (its title and description), which will be handled in a single call to the revision agent.
- QUESTION: No edits are needed to satisfy the request; the model will answer the user's question immediately.

Output from the revision agent follows a strict schema, and must consist of the following components: a *reply* which describes the suggested changes and a list of *actions* to execute along with the *lesson data* to update. Each action must be one of the types described in Table 5.1. The actions are then applied by the code, and the affected slides are shown one by one to the user for revision. Suggested changes are highlighted with a green bar, see Figure 5.9. An overview of the system can be seen in Figure 5.10.

Action	Description
updateLessonInfo	Change the lesson title and/or description.
updateSlide	Update a specific slide description, title, and/or image.
updateQuestion	Update a specific question and/or its answer.
addSlide	Add a new slide.
addQuestion	Add a new question.
deleteSlide	Delete a specific slide.
deleteQuestion	Delete a specific question.

Table 5.1: Available lesson editing actions. Each action is accompanied by lesson data specifying its parameters.

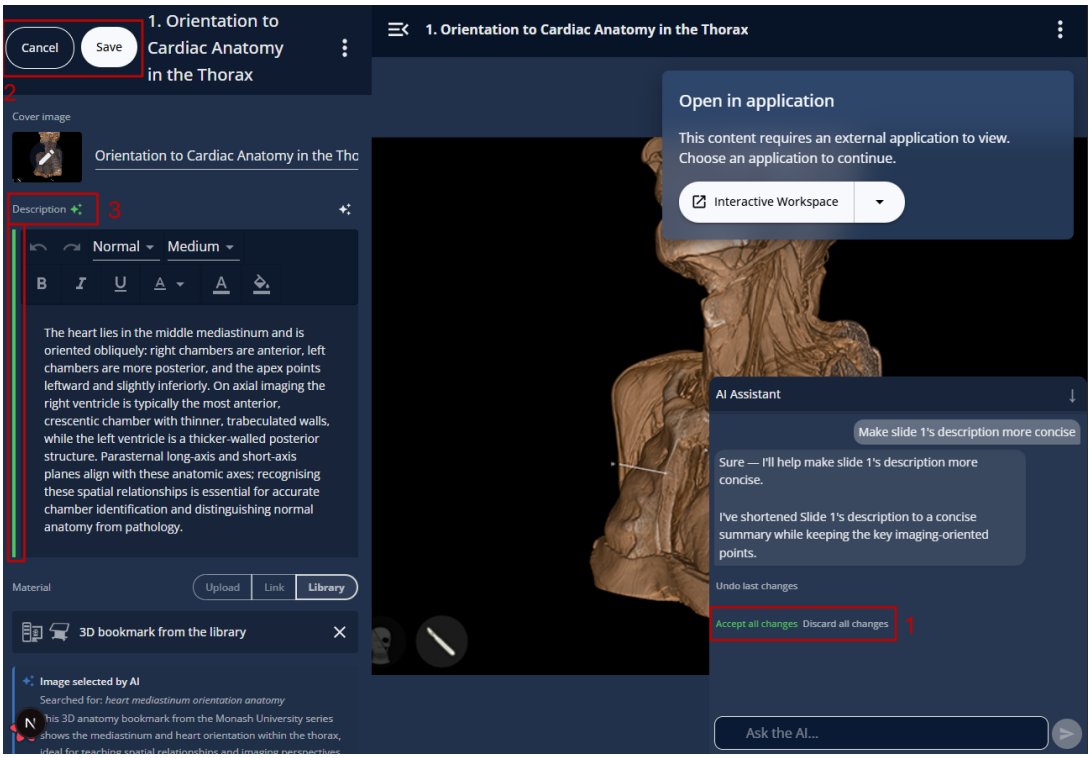


Figure 5.9: A snapshot of the lesson revision process, annotated with numbered red boxes. The agent has suggested changes that can be collectively accepted or discarded via the AI chat in the bottom-right corner (1), or slides can be handled individually via the Save and Close buttons in the upper-left corner (2). Fields changed by the AI (slide description) are highlighted with a green bar and a green sparkle icon (3).

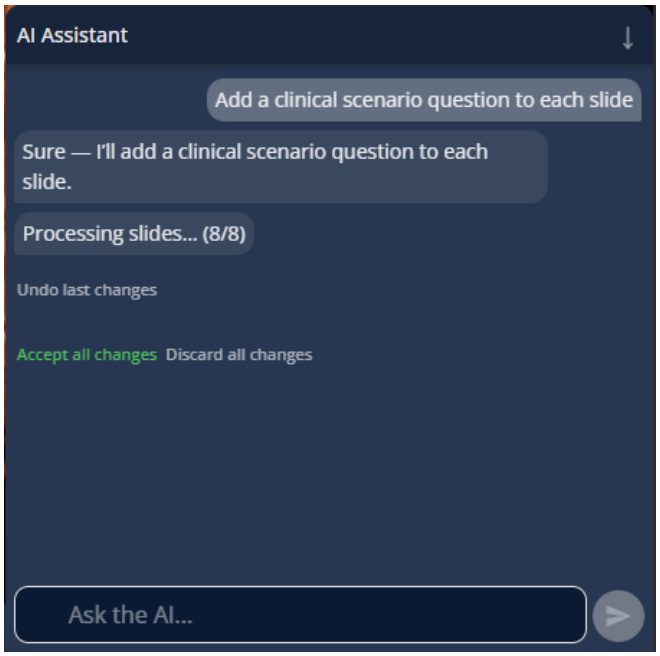


Figure 5.8: An example conversation in the AI chat. The agent has suggested changes that can be collectively accepted or discarded via the chat.

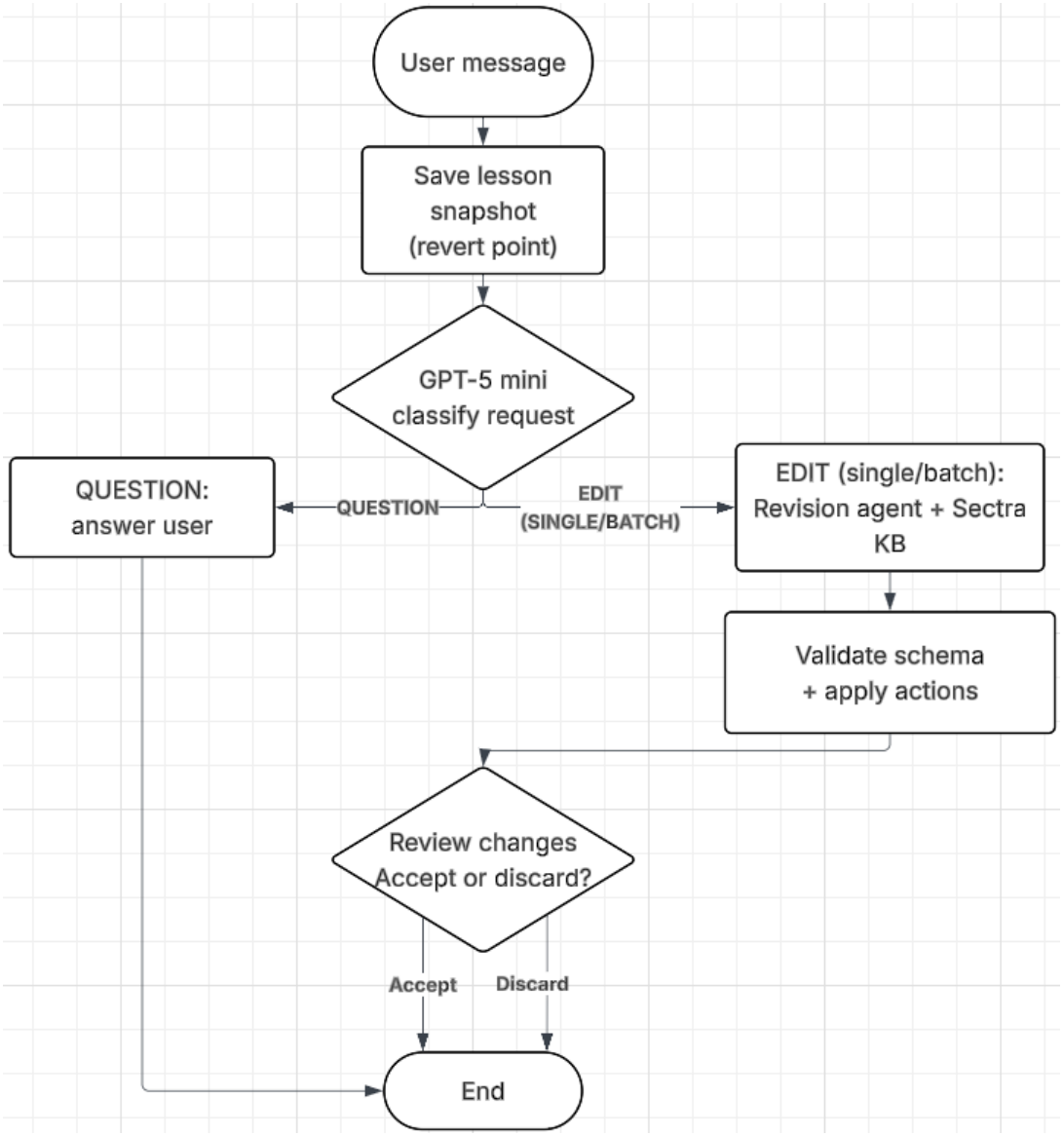


Figure 5.10: An overview of the AI Chat system. On each user message, the system saves a snapshot of the lesson and routes the request to GPT-5 mini to classify the request. If it is classified as a question, it is immediately answered. However, if it is an edit request, it is forwarded to a revision agent, which generates schema-validated edits. The resulting changes are applied and presented for user review. The process ends whether the user accepts or discards the suggested changes.

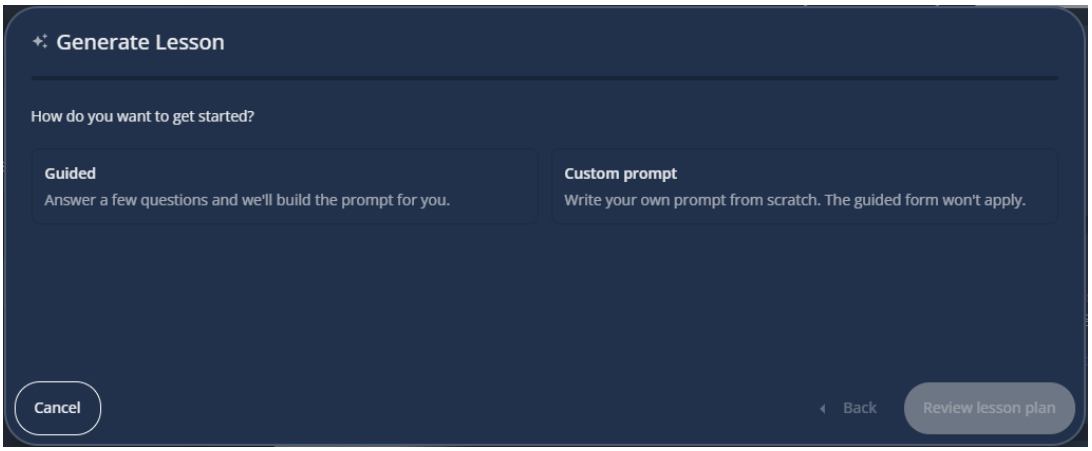


Figure 5.11: The first step of the lesson generator prompt form. The user can choose to write a custom prompt or fill in a form.

5.2.2 Updated prompt form

Feature description: The prompt form is an extension of the original prompt form (see Section 2.2.3). It includes the following (other than the input fields in the original prototype):

- a text field for the desired question style (for example, PBL or image interpretation),
- a drop-down menu of image types that may be included in the lesson,
- a text field for excluding topics from the lesson,
- buttons for selecting image focus (normal anatomy and/or pathology),
- examples of prompts and individual text field contents with varying levels of detail, and
- an option to write the prompt by hand, without using the guided form.

For some fields, the title was reworded, based on what users found confusing in the initial interviews: "Type of Lesson" was changed to "Intended use", "Number of Slides" was changed to "Lesson size", and "Number of Questions" was changed to "Question Preferences". Information about image availability depending on library content was also added. All parts of the updated prompt form can be seen in Figure 5.11, Figure 5.12, Figure 5.13, and Figure 5.14.

System details: The filled-in fields are used to construct a prompt, or the user's free-text prompt is copied directly.

5.2.3 Lesson plan

Feature description: The lesson plan is shown to the user after the prompt has been completed, as the last step before a lesson is generated. The plan shows the topic of each slide, along with a brief explanation of the content and the number of questions for each slide (see Figure 5.15). The user can leave revision suggestions and generate a new lesson plan, or generate a full lesson from the existing plan. The user can also return to the form to adjust parameters before generating a new plan.



Figure 5.12: The second step of the lesson generator prompt form, if the user chooses to enter a custom prompt. After writing the prompt, the user can generate a lesson plan, see Figure 5.15.

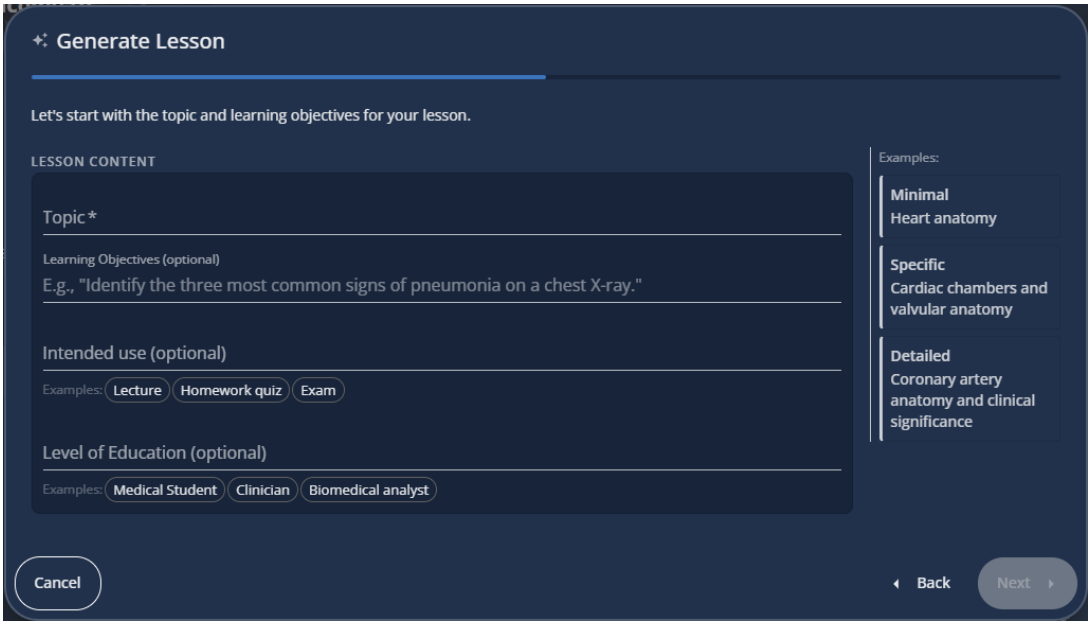


Figure 5.13: The second step of the lesson generator prompt form, if the user chooses to fill in the prompt form. The lesson topic is the only required field. On the right-hand side, clickable examples are displayed. There are also clickable examples under several of the text fields.

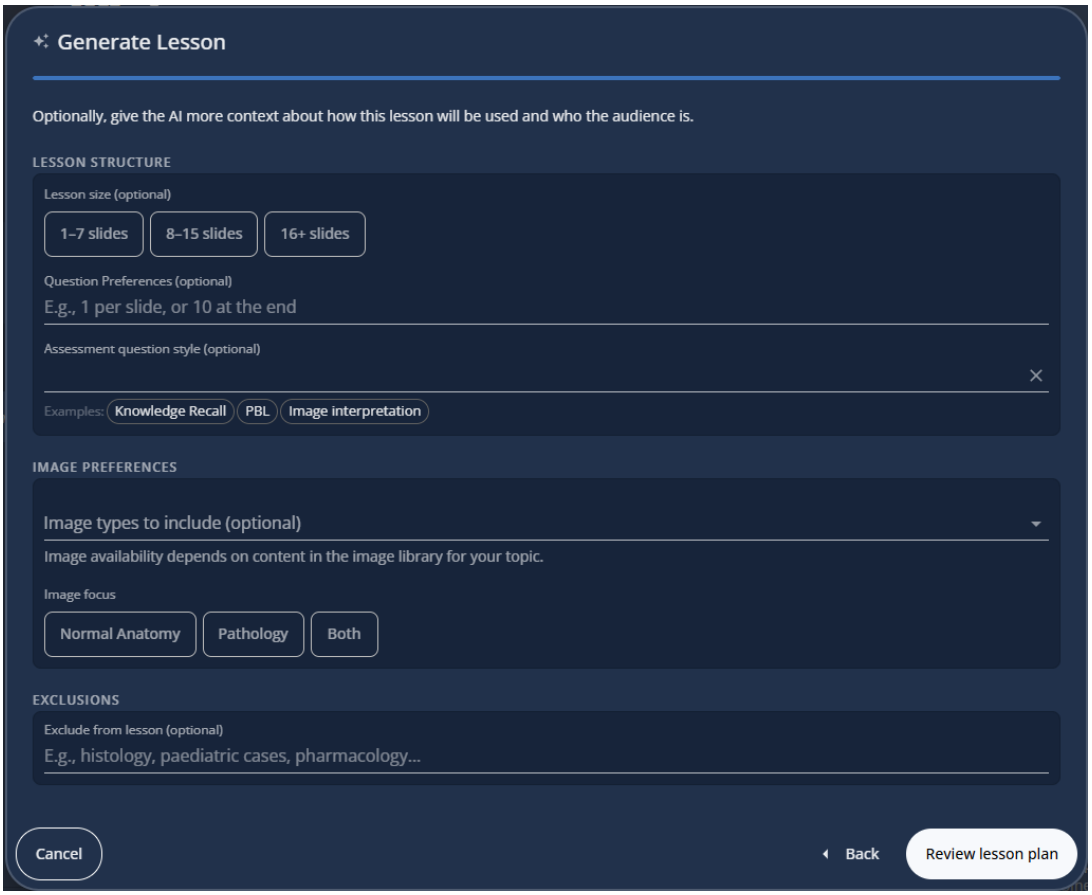


Figure 5.14: The third step of the lesson generator prompt form, if the user chooses to fill in the prompt form. All fields are optional. After this step, the user can generate a lesson plan, see Figure 5.15. In the image preference section, there is a disclaimer about image availability.

System details: The user's prompt is sent to a large language model (LLM), which returns a lesson outline. The structure is displayed to the user, who can provide feedback and regenerate the plan. On regeneration, the original prompt, the current plan state, and the user's feedback are sent to the LLM again. A revised lesson plan is generated and displayed to the user.

5.2.4 AI provenance banner

Feature description: On an AI-generated lesson slide, an informative banner is shown to the user (see Figure 5.16). The banner displays, other than the fact that the image was selected by AI, the search terms used, and the reason that the image was deemed relevant. There is also a clickable link directly to the image source, which shows the full link on hover.

System details: When an image URL is inserted via an AI-suggested action, the search terms and reason for choosing the image are both saved (these are read from the agent's output). When displaying a lesson, the provenance banner is shown if there exist saved search terms and selection reasoning for that particular image.

5.2.5 Batch-based lesson generation

Feature description: This change consisted exclusively of backend changes and was not visible to the users via the interface.

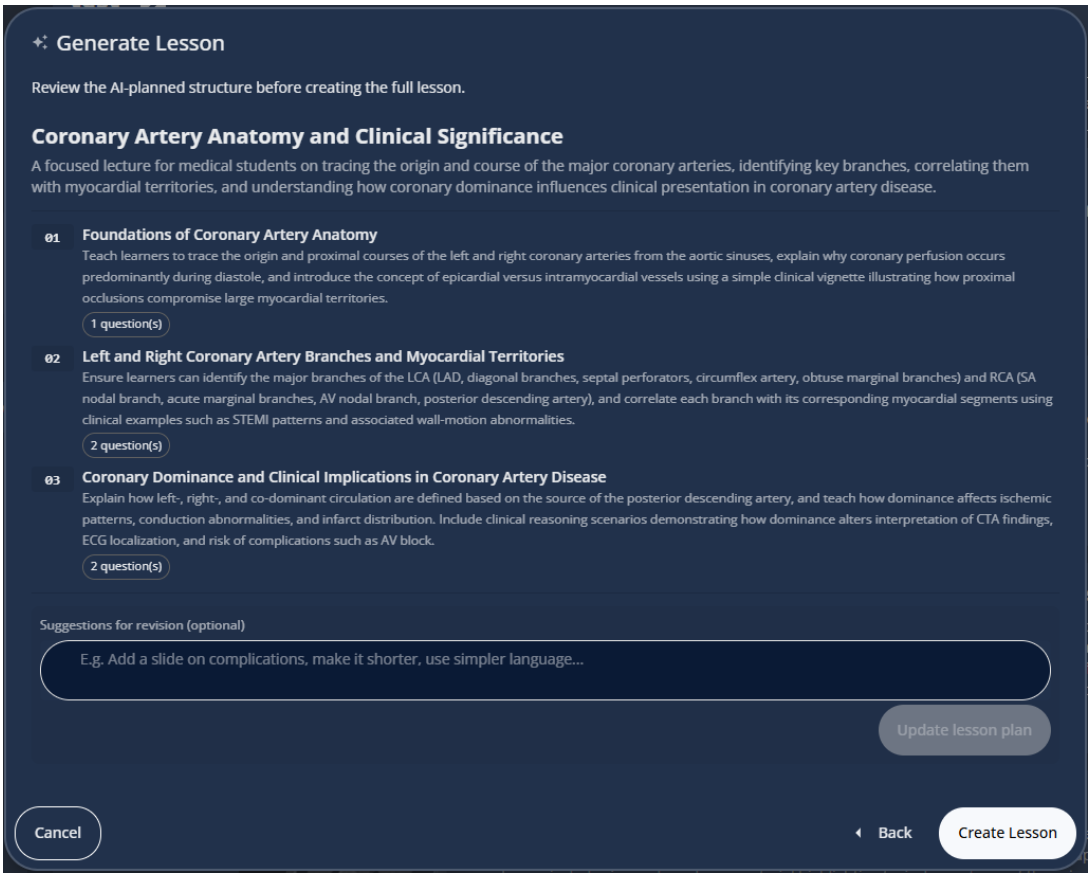


Figure 5.15: An example lesson plan. This is an outline of what the generated lesson will look like. The user can provide feedback on the lesson plan in the text field below it to generate a new version. The bottom right button initiates lesson creation based on the current outline.

System details: The generation technique was modified to produce slides concurrently in batches rather than generating the entire lesson in a single model response. The planned slides are divided into batches of four. For each batch, a single call is made to the retrieval agent, which has access to the medical image knowledge base, to obtain image URLs for that batch's slides; at most two such calls run concurrently, to avoid overloading the knowledge base. The batch size of four slides was determined empirically: larger batches caused the knowledge base's rate limits to be exceeded, so the size was reduced until generation completed reliably, with an additional margin.

The slides' text content does not require knowledge-base access and is therefore generated by direct calls to the language model rather than the agent, which lowers the risk of hitting rate limits and allows the text for all slides to be generated concurrently. Splitting the lesson across multiple requests gives each batch its own response budget, supporting more detailed per-slide content, while issuing the requests concurrently reduces total generation time.

5.3 Questionnaire results

In total, 15 users participated in the evaluation described in Section 4.3. For each prototype, the mean score of each construct (perceived usefulness, perceived ease of use, trust, and need to verify output) in the questionnaire is presented in Table 5.2. As seen in the table, Prototype A outperformed Prototype B on perceived usefulness ($M_A = 2.26, M_B = 3.21$), perceived ease of use ($M_A = 2.43, M_B = 3.42$), and trust ($M_A = 2.98, M_B = 3.56$). However, the total score for the need to verify output was identical between the prototypes ($M = 5.07$).



Figure 5.16: The red square marks the banner stating that the image in this slide has been selected by AI, as well as the search terms used and the reason the image was chosen.

Perceived usefulness was the strongest construct for both A and B (i.e., it had the lowest mean score). Trust and the need to verify output, on the other hand, had the least positive scores.

The mean score per item is presented in Table 5.3. A outperformed B on every item, except "I would need to carefully verify X's output", where the score was identical ($M = 5.07$). Excluding this item, which was not part of the S-TIAS scale, the weakest items for A were "I can trust Prototype A" ($M = 3.00$) and "I am confident in Prototype A" ($M = 3.00$). However, the weakest item for Prototype B was "I would find Prototype B to be flexible to interact with", which reached the neutral midpoint ($M = 4.00$).

The strongest items for A and B were "I would find Prototype A useful in my job," ($M = 1.93$) and "Learning to operate Prototype B would be easy for me," ($M = 2.73$) respectively.

5.4 Interview results

Follow-up interviews were held after the participants had completed the questionnaire. The interviews had three main focus areas: perceived usefulness, perceived ease of use, and trust. The findings from the interviews, regarding each of these focus areas, are described below.

5.4.1 Perceived usefulness

One common sentiment was that the lesson generator tool would reduce the time spent creating new lessons, but not eliminate it, as users would still need to review and revise the lesson.

Table 5.2: Mean score (M) and standard deviation (SD) per construct for the extended prototype and the original prototype (N=15). Scale: 1 = most positive, 7 = most negative.

	Extended prototype		**Original prototype**	
Construct	*M*	*SD*	*M*	*SD*
Perceived Usefulness	2.26	1.51	3.21	1.65
Perceived Ease of Use	2.43	0.82	3.42	0.81
Trust	2.98	1.14	3.56	1.60
Need to Verify Output †	5.07	0.88	5.07	0.96

† This item was originally reverse-scored: a lower score indicated a greater need for verification. In this table, the value has been reversed so that higher values indicate a greater need to verify output (negative), to enable consistency with the scale direction of the other items.

Table 5.3: Item-level mean scores for Prototype A (extended prototype) and Prototype B (original prototype) (N=15). Scale: 1 = most positive, 7 = most negative. Perceived Usefulness items are abbreviated; full wording is prefixed with "Using Prototype X in my job would…".

Item	**A**	**B**
Perceived Usefulness		
…accomplish tasks more quickly	2.20	2.87
…improve my job performance	2.40	3.60
…increase my productivity	2.40	3.07
…enhance my effectiveness	2.47	3.20
…make it easier to do my job	2.13	3.60
…be useful in my job	1.93	2.93
Perceived Ease of Use		
Easy to learn	2.27	2.73
Easy to get it to do what I want	2.47	3.80
Clear and understandable	2.80	3.33
Flexible to interact with	2.40	4.00
Easy to become skillful at using	2.27	3.33
Easy to use	2.40	3.33
Trust		
I can trust Prototype X	3.00	3.60
Prototype X is reliable	2.93	3.53
I am confident in Prototype X	3.00	3.53
I would need to carefully verify X's output †	5.07	5.07

† This item was originally reverse-scored: a lower score indicated a greater need for verification. In this table, the value has been reversed so that higher values indicate a greater need to verify output (negative), to enable consistency with the scale direction of the other items.

"We'll definitely save time, especially if you're starting from scratch and want to build something from the ground up. It will still involve some editing work, but it's nothing compared to creating everything yourself from the very beginning," said P4. P8 reiterated the sentiment: "I think as far as getting a first step out the door, and getting a bunch of information that's really complex gathered in one place already for you, I think that was a huge help. Then it's just a matter of editing it to make it the product you want it to be. So I think it was great, it would be a huge time saver." Another positive aspect brought up was reduced cognitive labor: "That sort of...white blank page mental block that you get when you've got to make something from scratch is removed, and you're very quickly into editing an advanced draft," said P11.

A recurring limitation was the size of the image library. P15 said that "The image data it had to choose from was extremely limited," and explained why having an extensive image library was important: "That's what you learn from: what is the difference between this image and that image?" P15 noted that if the system could retrieve multiple relevant images in seconds, it would be enormously time-saving compared to manually searching through the image archive.

Another participant, P5, also experienced limitations related to the image library when generating a lesson: "I tried to create a lesson on head traumas. There's basically only one case in the library, and it used that case for all my slides even though I specifically said 'find a subdural hematoma, find an epidural hematoma.' But they aren't in the database, so of course it couldn't find them." P11 corroborated the need for more images to search from: "I think being able to integrate it with my own case library and my own content library. That would be the really big thing."

The preference for Prototype A was explained by P1: "I experienced A as more powerful. It produced more advanced text, more information in each slide. And the AI chat was also very good to have as a complement. Those are really the main differences, I think." P8 similarly noted: "The first one, A, was very detailed. B was much shorter with too little detail. I would not have used the outputs of B; it ended up being so short that it limited its usefulness. A was very detailed, which can also limit usefulness, but when you're putting together lectures, I think it's nice."

The AI chat was a common favorite of many participants, for example, P9 said, "I would definitely use it, especially the chat option. I tried adding content through it, like 'I'm missing this type of thing, let's quickly add this slide.' If Prototype A were available, I would actually use it in my day-to-day."

Another advantage of Prototype A was the option to either select a guided generation process or to freehand a prompt, according to P5: "What I like about Prototype A is that you have these two options: guided or prompt. The guided option is easier than having to write everything yourself. It's really nice to have both, because if you're in a hurry, just use the guided and make a lesson. But with the prompt, you can go more in depth regarding what you want from your lesson, so it's more useful that way long-term."

The pedagogical qualities of the lessons were also raised by various participants. P2, for example, said, "If you think about it as self-study material, you'd need to develop the image descriptions a bit more — what is it I'm actually seeing here? It could be developed further to reach the level our students are at." P3 noted, "It was a bit simplified, for example, regarding melanoma, the truth is there are different types with different mutations. It wasn't wrong, but very simplified." P3 also noted a difference between the two prototypes: "On B, I didn't think it was as good. But on A, quite a lot of the text could actually have been used in a lecture."

The output quality of Prototype A can suffer from its verbosity, according to P8: "The output was a bit verbose, and I think it would, if kept in that format, lose some audience members. It would have to be shortened so that you would not lose attention."

However, not all participants noticed a difference in output quality between the prototypes, as P9 said, "Prototype A was more convenient to use to mold my lessons to what I want. I didn't necessarily pick up on any difference in output."

5.4.2 Perceived ease of use

The lesson generation process was generally perceived as easy, though some participants noted areas of confusion. For example, P8 said, "I thought it was very easy and intuitive. For those who procrastinate or put things off because the task feels so large, this takes away that burden. It makes it so easy and approachable that having to put together a lecture becomes less of a problem." P12 echoed a positive sentiment: "The way you're guided to build up a prompt was really clever. You could answer in segments rather than having to compose everything at once. The structure helped me focus on what was important, and there was

enough flexibility to be as detailed as I wanted." The flexibility to write a freehand prompt was also appreciated by P4, for example: "I used ChatGPT to help craft a prompt that I then adapted myself. I liked the freehand prompting, I thought it produced what I wanted just as well as if I had followed the guided form." P4 also said, "It was straightforward — there was nowhere I wondered how to proceed, or where I did something wrong and had to go back."

However, P10 mentioned a disadvantage of Prototype B: "In B I got a little lost — 'okay, now I need to select that, but what do they mean exactly?' In A, it was much easier and more intuitive to communicate what I wanted." P11 also mentioned not fully understanding the meaning of different form fields: "The descriptions were quite broad: ' early career or experienced': does that mean an experienced respiratory physician, a radiologist, a histopathologist? There's going to be a lot of nuance there." P6 expressed that while Sectra Education Portal isn't tailored for their particular professional field, Prototype A made it easier to adapt the content to what they teach.

Several participants also brought up the AI chat as a convenient tool to facilitate lesson revisions. P12 said, "With the AI chat, I could say 'change the cover image to a histological image rather than an anatomical one' — and it did it. With Prototype B, that was much more difficult; I had to go in and manually figure out how to change the cover image myself." P6 noted that Prototype A allowed for iterative refinement through conversation with the chatbot, whereas Prototype B required knowing exactly what you wanted from the outset. P1 similarly stated, "The biggest difference I experience is that you get a much more direct sense of control over the material, something you can change in multiple ways. If you can address every part through the chat, that's a very good complement. You might start by making changes there to adjust the style or lengthen questions, and then switch to manual editing for the fine-tuning." P7 reiterated that opinion: "I like that you can just chat and ask it to make things shorter, that's how I work myself." On the other hand, P9 highlighted a weakness of Prototype A: "Once you hit the edit space, I felt like I couldn't go back and forth between edits very easily."

5.4.3 Trust

Multiple users pointed out the need to verify the generated material, regardless of which prototype was used. For example, P9 said, "I have a general distrust in AI, so I always need to review the work they give me." P11 also underlined the need for review of AI-generated lessons: "I don't think it's contradictory to say I have a pretty high level of trust in the quality, but at the same time I'm going to check everything, just to be able to relate to the trainees." P5 similarly stated, "I would have to use it for a while to see if something pops up. As of now, I would review it manually to see if I can, you know, vouch for it." Some participants also highlight manual review as a professional duty: "If my name's on it, I would want to edit it and make sure it's something I'm okay with," said P8.

To decrease the need for manual review, P14 said that having references in the text might be beneficial: "If I knew where the information came from, I wouldn't need to worry in the same way or scrutinize it as hard." The subject was also mentioned by P8, who said, "It would be good to have sources. If something seemed wrong, that's where references could help, because then you could dig down." Additionally, P3 mentioned that references in the generated text would make them trust it more, and P10 agreed, "If I can be sure that there is some strong reference behind, I will still check it, but I will feel more confident at least." The RAG system design also had implications for trust for some of the participants: "Since it draws from the material in the system, I feel there's a quality assurance built into what's presented," said P2.

Two participants brought up trust calibration over time. One of them was P6, who said that working with the tool over time made them less rigorous about reviewing the output, as familiarity can lead to overlooking errors. Furthermore, P4 stated, "I would have reviewed it manually. But I think that's just how I am as a person — the first things I do, I want to check that they're actually correct. But after doing two or three lessons and seeing that it works, I

would probably be able to start using them directly with students. I would always note somewhere that AI was involved in creating the material, though." Although in Prototype A, there was a disclaimer that images were AI-selected, P4 pointed out that they wanted a disclaimer for the text as well: "It's just as important that the text is also marked as AI-generated. I think it should be there, because it's a source of information for whoever consumes the material I've created that AI was involved in producing it."

Trust was higher for Prototype A than for Prototype B, both in the questionnaire and in the interviews. "I felt like prototype B was less accurate", said P8. P4 agreed with this: "With Prototype A, I could trust it more. It was more comprehensive and felt like a better foundation." P5 stated, "Prototype B doesn't always do what it says."

P13 brought up the accuracy of the generated material as a flaw: "It placed a bit too much emphasis on certain aspects that aren't really clinically relevant." P12, on the other hand, reported a more positive experience: "There were no AI hallucinations. It never gave the wrong answer, which gave me more confidence that it was working as intended."

6 Discussion

In this chapter, the limitations of the study are discussed, covering the size of the knowledge base, the participant group, and the participants' limited exposure to the prototypes. Then, the results of the project are analyzed with regard to the research questions in Section 1.3. In the next section, the method of the project is criticized and discussed, analyzing topics such as participant selection, the structure of the pre-study, and more. Lastly, the project is put in a wider context, examining its environmental and social effects.

6.1 Results

The implications of the questionnaire and interview results are discussed below, based on the research questions presented in Section 1.3.

6.1.1 Extending the prototype to support perceived usefulness, ease of use, and trust

As described in Section 4.2, the system design decisions were largely based on the findings from the pre-study interviews. Participants frequently suggested features themselves, and many of these suggestions aligned with the established design principles presented in Section 3.2. The evaluation results indicate that several of these design decisions were meaningful to users.

Perceived usefulness: The extended prototype scored notably better than the original prototype on usefulness (M=2.26 vs M=3.21), and the interview data suggests the AI Chat was a primary driver of this difference. This aligns closely with the findings from the pre-study, as described in Section 5.1, where iterative AI-assisted revision was both the most frequently requested feature and strongly supported by prior research on human oversight, see Weisz et al. [15], and collaboration in generative AI systems, see Lin et al. [16]. Participants appreciated being able to iteratively refine lessons via the chat, as it increased their sense of control over the generated content and allowed them to progressively improve outputs rather than restarting the generation process. P1 stated that "The biggest difference I experience is that you get a much more direct sense of control over the material, something you can change

in multiple ways," and P9 said they would "definitely use it, especially the chat option." Notably, one participant who tested the extended prototype first even interpreted the absence of AI-supported lesson revision in the original prototype as a system error rather than an intentional design difference. This implies that conversational AI-assisted revision is an effective extension for improving the perceived usefulness of AI-generated lessons.

The results of introducing batch-based lesson generation were mixed: some did not perceive any meaningful difference in the generated lessons, while some expressed a clear preference for the output of the extended prototype. Among the latter, the lessons generated with Prototype A were often perceived as more detailed and informative, affecting the perceived usefulness positively. However, the perceived difference in output quality could also be a consequence of differing prompts, which in turn may be affected by changes made to the prompt form. Due to these uncertainties, it is not possible to determine whether the design rationale for batch-based lesson generation was supported by the results.

The preview of the lesson structure (the Lesson Plan) was not frequently noticed by users, based on the interviews conducted in the evaluation phase. This feature was added as a result of user requests in the initial interview round, as well as to facilitate human-AI collaboration, reduce system latency, and make invocation more efficient, which in turn was intended to increase the perceived usefulness. Although system latency was not as frequently brought up as an issue in the final interviews compared to the initial interviews, there did not seem to be a common notion of experiencing less system latency with Prototype A than with Prototype B. Possibly, system latency was more noticeable in the initial interviews as participants were being observed while testing the prototype. It is possible that the Lesson Plan was not noticeable to the users due to the implemented interface; for example, users may have been focused on the final generated lesson, rather than the intermediate planning step. It is also possible that the Lesson Plan was not distinct enough from the rest of the generation flow to be perceived as a separate feature. The limited effects could also be because the intended benefit of reduced latency could require repeated use to become perceptible. In conclusion, extensions targeting the generation process may require longer exposure to be meaningfully evaluated in a single-session study.

Perceived ease of use: The extended prototype outperformed the original prototype on perceived ease of use as well (M=2.43 vs M=3.42). The main system addition for this construct was the updated prompt form, which was notably overlooked in the interviews by most participants. When asked about differences in the generation process between the two prototypes, many said they did not notice any or did not have any preferences. However, some noted and appreciated that Prototype A allowed for freehand prompting as an alternative to using the guided form, with the guided form providing structure and reduced ambiguity in the prompting process, while freehand prompting offered greater flexibility for experienced users. Some also felt the form in Prototype B was more confusing or made it more difficult to tailor the lesson to their specific needs. A few times, the prompt examples were also brought up as a helpful aspect. This suggests that offering multiple interaction modes may better accommodate different user preferences and levels of prompting experience, and that the changes made could have a positive effect on perceived ease of use for a portion of users.

The limited attention given to the updated prompt form was noteworthy, as the updated form was supported both by design principles (help users effectively prompt, and provide examples of working prompts) and explicit requests from interview participants for prompt examples and input fields to add. Possibly, the changes made did not have the desired effect of making the prompting process easier and more intuitive. For example, the reworded fields could be equally ambiguous, or the added fields could be unsuccessful in making the user feel more guided. The prompt examples were placed on the right-hand side of the form, which could have resulted in them going largely unnoticed. Another possibility is that the

updated prompt form was too similar to the original version for the participants to notice a difference.

Beyond the prompt form, the AI Chat also contributed to perceived ease of use, as several users stated that it made lesson revision easier. For example, P12 said, "With the AI chat, I could say 'change the cover image to a histological image rather than an anatomical one', and it did it. With Prototype B, that was much more difficult; I had to go in and manually figure out how to change the cover image myself." In summary, the AI Chat improved ease of use by removing the need for users to know exactly what they wanted or how to achieve it manually, allowing them to refine the lesson through natural conversation instead.

Trust: Although the extended prototype outperformed the original on the trust construct ($M = 2.98$ vs $M = 3.56$), it was the weakest construct for both prototypes. Even though previous research by Weisz et al. [15] has established origin provenance as a design principle, the AI Provenance Banner, disclaiming that images were selected with AI, did not seem to affect users' perceptions much, as the feature was seldom mentioned by users. The interview data implies that the increased trust for the extended prototype was instead caused by the perceived improved output quality. For example, P4 stated, "With Prototype A, I could trust it more. It was more comprehensive and felt like a better foundation." Trusting the system with higher output quality more could be an example of appropriate trust, which is described by Lee and See [20]. The perceived increase in output quality was possibly due to the Batch-based Lesson Generation, as it allowed for more detailed slide contents. However, the improved quality could also be attributed to the Updated Prompt Form, resulting in more descriptive prompts. The lack of interest in the AI Provenance Banner could be a consequence of the banner not being visible enough to the users due to its design. Additionally, several users mentioned wanting references for the generated text and that this would increase their confidence in the output. This indicates that their trust concerns were broader than image provenance, and forms a direction for future development.

6.1.2 Educators' perceptions of AI-generated lessons

Overall, the evaluation results imply a positive perception of AI-generated lessons. Perceived usefulness was the strongest construct for both prototypes, signaling that educators would have use of an AI lesson generator tool in their work. Prototype A scored especially well across all three constructs: usefulness, ease of use, and trust. These results could indicate that educators may be realistically willing to integrate this tool into their workflow, since, according to TAM, perceived usefulness and perceived ease of use are strong predictors of technology adoption [23]. Although Prototype B scored positively as well, the results indicated a clear preference for Prototype A, suggesting that educators would rather use Prototype A in their workflows.

Despite trust being the weakest construct, the mean score implied that both prototypes were perceived as trustworthy by the participants. The finding that trust scored lower than usefulness could indicate that educators see practical value in the AI lesson generator, despite reservations about AI reliability. In practice, this could mean that educators might use the generator as a drafting tool, rather than an autonomous lesson generator.

Notably, both prototypes scored poorly on the need to verify, with an identical mean to one another, even though trust scores varied. This can be an indication that, for AI-generated materials, there is a weaker link between trust in the product and the need to verify its output. In the evaluation interviews, users stated a few different reasons for feeling the need to review the material: to be able to relate to the trainees, a sense of accountability for the content, and general distrust in AI. The latter reasons, in particular, are compatible with research by Ghimire et al. [30], who found that the main concerns among educators regarding generative AI included the risk of incorrect output. This suggests that educators are not willing to fully automate lesson creation, even if the tool might seem trustworthy, as the need to verify

might be driven not only by distrust in AI systems but also by professional accountability and ethical responsibility.

6.1.3 Factors influencing perceptions

The interviews provided qualitative depth that uncovered factors that could have influenced the questionnaire scores. One of these is the coverage of the underlying knowledge base, as mentioned in Section 6.3. Multiple users stated that the image library would have to be expanded to make the product truly useful to them. Furthermore, the RAG mechanism itself can instill confidence in users, according to some participant interviews. This aligns with design principles established by Weisz et al. [15], who recommend letting the user know the origins of the system's output. An explanation for this could be that users might trust systems more when they can infer where information comes from. This aligns with interview findings, as several participants requested references in the generated content. It is also supported by design principles about system transparency.

Another factor mentioned by several users was the qualities of the generated output: its verbosity, level of detail, and accuracy, for example. Participants mentioned that too little detail renders the generated material less useful, but very verbose output can also limit usefulness. A low level of detail can make the output less accurate, which in turn can affect its trustworthiness. These findings suggest that usefulness is not solely dependent on correctness, but also on whether the generated content matches educators' preferred teaching style and desired level of detail.

6.1.4 How feature-level enhancements affect perceptions

As shown in Section 5.3, Prototype A outperformed Prototype B by approximately 1 point across all three constructs. This implies that the enhancements in Prototype A positively affect all measured dimensions, thereby indicating that they increased educators' willingness to adopt the tool. Perceived usefulness and perceived ease of use have been identified by Taheri et al. [31] as influential factors for educators' willingness to adopt AI tools in the classroom. Since Prototype A scored especially well across all three constructs, these results could indicate that educators may be realistically willing to integrate the extended tool into their workflow [23]. As discussed in Section 6.1.1, however, not all individual features contributed equally to this outcome: the AI chat was the most impactful, while features such as the lesson plan preview and AI Provenance Banner went largely unnoticed.

6.2 Method

This section focuses on the decisions made regarding the method of this thesis. The project's replicability, reliability, and validity are considered, and the reasoning behind all phases of the project is explained.

6.2.1 Participant selection

Participant selection is described in Section 4.3.1 and was carried out to represent the intended users of the system: educators already using or wanting to use Sectra Education Portal. However, whether this selection was actually representative of the user base is not guaranteed, as there is no clear mapping of user data, such as their professions, experience in Sectra Education Portal, and areas of expertise. Instead, the user base composition has been estimated with the help of Sectra employees in the Medical Education department. Consequently, the results of this study may not accurately represent the opinions of the full user base.

The participant group contained a mix of non-educators and educators with varying levels of experience in Sectra Education Portal, which contributed to a range of perspectives on the prototypes. However, the sample was still predominantly composed of users within medical education, particularly radiology, which may limit the generalizability to other medical areas or educational domains. At the same time, this bias was intentional, as the system's knowledge base primarily covers radiology and histology content. Selecting participants from these areas, therefore, allowed for a more meaningful and fair evaluation of the generated outputs.

A small subset of the participants were Sectra employees who do not teach. They differ from the rest of the participant group, as they do not directly use Sectra Education Portal in teaching. Therefore, their perspectives might not fully reflect those of educator participants. However, since the Sectra employees in this study work closely with educators and are familiar with their needs through their professional roles, their input was still considered valuable as a complementary perspective.

Five out of six participants from the initial interview round partook in the final evaluation. While this allowed for deeper insight into how user opinions evolved over iterations of the lesson generator tool, it also introduced a risk of bias. This is because recurring participants were exposed to the original prototype (Prototype B) twice, whereas they only tested the extended prototype once. This could potentially lead to them becoming more comfortable with Prototype B and favoring it because of this comfort level. Consequently, this could skew questionnaire scores for Prototype B.

6.2.2 Pre-study

The purpose of the pre-study was to find direction in how to extend the prototype; The literature review, in combination with the user interviews, was intended to provide a base for system design decisions. The implemented system changes were decided based on established design principles, frequent mentions in user interviews, and estimations of what could realistically be completed within the scope of this project.

The small number of participants (N=6) in the initial interviews means that results should be interpreted as indicative and not representative. The focus on radiology and histopathology educators could also have shaped the design decisions toward a system optimized for this particular medical area, which could make the extended prototype less useful to educators within other medical areas.

The analysis of the initial interview data was conducted by a single researcher, meaning that interpretations are subjective and made from a single perspective. Groupings and themes were created by the same single researcher, once again affecting the reliability of the study negatively [38].

6.2.3 Implementation

As described, feature selection for implementation was based on pre-study findings and design principles. The evaluation results are consistent with this approach being sound, as all resulting construct scores were higher for the extended prototype. The AI Chat, in particular, was strongly supported by both user needs and previous literature and had a significant impact on the final result. However, a clear limitation of this method is the small number of participants in the initial interview round (N=6). Consequently, the design decisions could be skewed toward a narrow set of needs, as features not mentioned by participants were largely overlooked. The focus on radiology and histopathology educators in the pre-study may have further skewed feature selection toward the needs of those domains.

The feature design was grounded in user opinions rather than a purely heuristic process, to reduce the risk of implementing unwanted features. A purely user-driven approach was

also avoided, however, as it risked producing requests that were out of scope or not generalizable. Therefore, a combination of the two was preferred.

Although the general results reflected the intended effects of the implemented features, some features had a limited impact, based on the user interviews. The lesson plan preview and the AI Provenance Banner, for instance, went largely unnoticed by participants despite being grounded in both user requests and design principles. This may suggest that user-stated preferences do not always translate to noticeable improvements in practice. It is also possible that the limited effects of these features are partly attributable to the small sample size in the final evaluation.

During the design and implementation of the features, Medical Education team members were consulted repeatedly. This helped ensure that decisions were informed by domain knowledge. However, it also introduces a risk of bias, as it may have led to the solution being optimized for Sectra's internal personnel rather than end users. Additionally, the same researcher was responsible for conducting the pre-study and implementing the features. This means that the prioritization of needs was shaped by a single perspective. Consequently, the interpretation of the pre-study results and the implementation results were not independently verified.

Some requested features could not be implemented due to the limited scope of this project. This means that the extended prototype does not represent the full set of enhancements identified as potentially useful, and that the evaluation reflects only a subset of the improvements that could be made to the system.

6.2.4 Evaluation frameworks

This project was focused on uncovering users' perceptions and opinions of an AI lesson generator tool, rather than the tool's technical correctness and performance. The goal was to assess factors such as usability, trust, acceptance, and attitudes toward AI-generated educational content. To increase the validity and comparability of this study with previous research, a few different existing frameworks were considered. The Technology Acceptance Model (TAM), described in Section 3.3.1, is a framework for assessing users' acceptance of technology. TAM is widely used, including in educational technology and research on AI-assisted systems, for example, in some of the research handled in Section 3.5.1. As TAM is focused on perceived usefulness and perceived ease of use as predictors of acceptance, this framework was deemed suitable to discover users' opinions on whether the lesson-generation system is useful in their workflow, whether the interface was easy to work with, and, by extension, whether they would realistically adopt the system in practice. Because this project evaluates a human-facing tool rather than backend algorithms, TAM was especially suitable.

An AI lesson-generation tool introduces other concerns beyond ordinary usability, such as trust and autonomy. TAM alone does not capture these attitudes, which is why another framework was chosen: the Short Trust in Automation Scale (S-TIAS). S-TIAS, described in Section 3.3.5, is a framework for assessing trust, which has been validated for use in AI contexts. This framework complements TAM in this study by focusing on perceptions of intelligent systems, providing a more holistic evaluation of the system. However, generative AI is known to introduce false information in generated materials, so-called hallucinations [5]. This phenomenon creates a potential need to verify the output of a generative AI system, but user attitudes regarding this were not captured by any of the selected frameworks. Therefore, a fourth trust-related questionnaire item was added, see Section 4.3, which assesses the perceived need to verify the generated materials. The item was formulated for this study, and its wording ("I would need to carefully verify the prototype's output") was kept deliberately simple and was aligned with the response format of the S-TIAS items so that it could be administered alongside them on the same seven-point scale.

The strengths of the chosen frameworks are that they are established and validated in prior research. They are both suitable for questionnaire-based studies and provide quantifiable results that can be used for comparison between two systems. However, self-reported questionnaires may introduce biases, as the subjects are stating their own perceptions and attitudes rather than demonstrating actual behavior [39]. As the fourth trust-related questionnaire item was not originally part of either framework, it has not previously been validated in this context. Although this introduces some uncertainty about validity, it was deemed an important aspect of this evaluation, and the question was considered sufficiently straightforward and easy to interpret.

Previously, TAM has been extended for multiple different contexts, but in this project, the standard form was used, which may oversimplify technology adoption. Another limitation is that S-TIAS measures attitudes and not long-term behavior. This means that users' behaviors in practice may not reflect the results of this study.

Alternative methods considered were, for example, evaluation based purely on usability heuristics, or qualitative methods only. However, the first option would not capture users' attitudes, which was the primary focus of this thesis, and the latter option would be harder to quantify and compare. In summary, the combination of the aforementioned frameworks was deemed appropriate because the project evaluated an AI-supported educational tool where both usability and attitudes toward AI were central to understanding user acceptance.

6.2.5 Evaluation procedure

The purpose of the user tests was to provide the users with a basis for the following prototype evaluation, to compare users' perceptions of Prototype A and Prototype B regarding usefulness, ease of use, and trustworthiness. About half (N=8) of the participants tested the extended prototype before the original, while the other participants (N=7) followed the opposite order, to reduce the impact of any bias stemming from which prototype users tried first. According to the results, participants who tried the original prototype second gave it a better score than the participants trying that same prototype first. The same was true for the extended prototype. This implies that the counterbalancing was a justified precaution. However, these results should be seen as indicative rather than conclusive, given the small sample size.

The participants were exposed to the two different prototypes over a limited time period. As mentioned in Section 5.4.3, users' perceptions of the prototype may vary over time, meaning that the scores given by the participants could change with more usage. In the initial interview round, several users mentioned finding it difficult to evaluate a tool after only being able to try it in a short interview session with the researcher present. To counteract this, the user tests in the evaluation phase were designed to give participants more exposure to the prototypes. However, as mentioned in Chapter 1, educators have limited time, and it was deemed too difficult to schedule long testing sessions with the researcher present. Therefore, participants were instead given a set of tasks to complete at their own convenience in an unmoderated setting, meaning that the participants completed tasks without observation. Consequently, it was not possible to control what participants actually attempted or whether any technical issues went unreported. This limitation could have been addressed by screen recordings of participants. However, screen recordings could have exposed sensitive or identifiable content on participants' computers, raising additional consent and data-protection requirements. It could also increase the participant burden, which might have deterred participants. Furthermore, the focus of this study was on participants' perceptions rather than their task behavior, meaning that screen recording was less central for this study.

The tests were designed with the purpose of simulating realistic usage: generating a lesson within the participant's area of expertise, revising it, and then using acquired knowledge to generate a new lesson or further improve the existing one. The expected benefit of this approach was that participants would be able to use their domain knowledge to evaluate the

quality of the generated materials. However, this also means that all participants reviewed different materials with varying levels of quality, accuracy, and relevance. This negatively affects the reliability and comparability of the study. At the same time, the structured nature of the tasks may also reduce the realism of the evaluation, as participants were instructed on what to do rather than interacting with the system entirely naturally.

The window for completing the tasks and participating in the final interview was intended to be three weeks, as it was estimated that participants would have time to finish the evaluation within this period and that it would be possible to conduct 15 interviews during that timeframe. However, the period was extended by three days, as some participants could not find time to complete the evaluation within the planned window. Some users also scheduled their interviews one to two weeks after testing the prototypes due to limited availability. One potential consequence of this is recall bias, as participants may misremember or forget details of their experiences. This was observable in several interviews, where participants occasionally responded with "I can't remember" when asked about specific aspects of their experience. This may have affected the validity of the interview data used to contextualize the questionnaire results. Furthermore, some interview transcripts were of limited quality due to automatic transcription and language differences, which may have resulted in loss of nuance in some participants' responses.

Interviews were selected as a qualitative method to understand user attitudes and the reasoning behind these attitudes. Compared to methods such as observation, interviews allowed for follow-up questions to gain deeper insight into what affected participants' perceived usefulness, ease of use, and trust. Focus groups were eliminated due to the risk of participants influencing each other's opinions, but also because of logistical reasons, as participants had busy schedules and were spread out across different time zones. For this project, interviews were therefore deemed the most appropriate.

6.3 Limitations

One of the main limiting factors of this research is the quality and coverage of the underlying knowledge base, as seen in Section 5.4. This affected several participants' experiences and might have influenced their questionnaire ratings. That said, the extent to which the underlying data affected the collective user experience is itself a noteworthy finding, suggesting that infrastructure quality can be a factor in the adoption of AI-generated lesson tools.

Another significant limitation is the limited number of participants (N=15). A small sample size means that any potential noise may have a significant impact on the total result. For example, there was a clear outlier among the participants; one participant gave consistently negative scores for both prototypes, especially across perceived usefulness and trust. This is reflected in the standard deviations seen in Table 5.2, and means that conclusions about the typical user experience should be interpreted with caution. When excluding the outlier's questionnaire response, the resulting scores are marginally more positive for both prototypes. However, the extended prototype still outperforms the original by approximately one point across all constructs.

A small data set could also increase the risk of bias and limited representation. One observable phenomenon from the questionnaire results (see Section 5.3) was a high standard deviation; in other words, there was a large variability in the small data set. This makes it harder to draw firm conclusions about the typical user experience.

Lastly, the evaluated tools are immature prototypes, meaning that they are not yet ready for commercial use. Immature products may suffer from software bugs and limited resources, among other limitations. Because neither Prototype A nor Prototype B is a finished product, the evaluation can only be seen as indicative of user opinions of a mature AI lesson generator tool.

6.4 Sources

Most sources used in this thesis are peer-reviewed scientific articles, books, and conference proceedings. Older, foundational research has been used in combination with recent research in specific areas. This can result in a deeper understanding of the state of the research, but using older papers may also introduce concepts or assumptions that do not fully reflect modern generative AI systems. At the same time, very recent research may not yet be fully validated through long-term use. Using papers from both categories was a deliberate decision to gain more perspectives on appropriate methods and background information for this thesis.

The most central sources for this project concern technology adoption, trust in automation, and design-oriented research on human-AI interaction. Because generative AI is a rapidly evolving field, some findings and recommendations may change as the research matures. Furthermore, some related work included in this thesis originates from broader educational or generative AI contexts, which may limit direct comparability to the system evaluated in this study.

6.5 The work in a wider context

In this chapter, the effects of the project on a larger scale are considered. First, the environmental effects of generative AI are regarded, including energy and fresh water consumption. Next, potential social effects, such as work displacement and overreliance on AI, are discussed.

6.5.1 Environmental effects

The environmental consequences of generative AI's emergence are a highly relevant topic in current research. The exact effects have been proven hard to quantify, but recent work has suggested ways to estimate the environmental impact of generative AI, with Life Cycle Assessment (LCA) being a central method [40] [41]. LCA can be used to understand the environmental impacts of a product over its entire life cycle, and recent research by Berthelot et al. [42] using LCA has uncovered that generative AI services have a substantial impact on the depletion of minerals and metals, climate change, and energy usage. However, internet infrastructure and end-user terminals (such as laptops or smartphones) have proven to constitute a majority of the impact on mineral and metal depletion (>85%), as well as a significant percentage of the energy usage (>30%) and contribution to climate change (45%). While generative AI may not directly cause end-users to buy new terminals, increased generative AI usage contributes to increased device and network energy consumption, according to Berthelot et al.

The effects of generative AI systems are not limited to mineral depletion, carbon emissions, and energy usage: for example, mining and producing the necessary components lead to local deforestation, water contamination, water pollution, and air pollution, say Hosseini et al. [40]. The same paper also states that a portion of the raw materials used for components in generative AI systems is non-recyclable, which poses a challenge for waste management.

The negative effects of generative AI should be carefully considered and weighed against potential benefits before deploying any such system, including the system constructed in this project. However, there may also be positive consequences of generative AI usage. Generative AI can be used to streamline essential parts of society, potentially resulting in resources saved. For example, being able to generate and revise lessons in a relatively short amount of time may result in educators spending less time manually producing equivalent materials. Although time saved cannot be equated to environmental savings, this could, in turn, result in a lower organizational resource usage over time. Facilitating online lessons could also promote remote learning and asynchronous teaching. Potential consequences of this include reduced commuting, which, for example, could decrease traffic emissions.

6.5.2 Social effects

A critical aspect of generative AI is the correctness of the generated materials. LLMs have been known to produce so-called hallucinations, i.e., false information [5] [43]. In healthcare, especially, misinformed healthcare personnel could have serious consequences for patient health. Therefore, it is imperative that AI-generated lessons in medical education are correct and accurate. Ensuring factually correct content is a challenge; as of today, there are no standardized frameworks for assessing hallucinations, and generalizability to different domains, for example, cannot be guaranteed for the hallucination detection methods that exist [43]. In other words, it is not only difficult to prevent hallucinations but also to identify them. Before a product like the AI lesson generator becomes commercially available, the risk of hallucinations must be thoroughly assessed, and mitigation strategies must be considered. Strategies for improving correctness in AI-generated materials include using RAG for the textual content as well, which would provide factual grounds for the lesson content. Another, related, mitigation technique is Knowledge-Augmentation, which entails that the model has access to structured knowledge that is injected when reasoning [44]. Another potential mitigation strategy could be human review and revision. However, this option raises new challenges, such as how to ensure that the generated content is thoroughly reviewed by a human with domain expertise.

Beyond factual correctness, AI-generated educational content also raises concerns regarding how educators interact with and rely on such systems. Although there are strategies to encourage reflection when AI-generating materials, such as introducing deliberate delays to prevent automatic behavior [15] [45], there is a significant risk of disproportionate trust in AI. This was indicated in the final user interviews, where some participants mentioned that they likely would review the generated materials less and less with time. Recent research suggests that overreliance on AI, other than the risk of accepting faulty information, can have personal effects, such as reduced critical thinking abilities [46]. There is also the ethical dilemma of whether extensive reliance on AI-generated educational material conflicts with principles of academic integrity and educator accountability.

In addition to hallucinations and inappropriate trust, generative AI systems may also reinforce existing societal and medical biases [47]. Bias propagation can happen when a model is trained on biased data, for example, which causes the system's output to reinforce that bias. This can potentially result in discriminatory or unfair content. In medical education, this could have a negative impact on healthcare equity.

One of the main issues that motivated this thesis was the heavy workload and limited time of educators. The tool was designed to facilitate and reduce the time needed to create educational materials. For educators, this product could potentially increase efficiency. In turn, this might contribute to reduced stress or better performance at their jobs. Another potential benefit of this project is democratization, as it lowers barriers for lesson creation for educators who are less comfortable with digital tools.

Replacing humans with AI in certain tasks raises concerns about job displacement, not exclusively in education. However, due to the concerns above, human oversight remains essential even when AI systems are used to support content generation, especially in safety-critical domains such as healthcare education.

7 Conclusion

The purpose of this study was to explore generative AI in the medical education domain by understanding educator perspectives on the ease of use, usefulness, and trustworthiness of AI-generated lessons. At the start of the project, Sectra already had an existing prototype. Enhancements to this prototype were designed and implemented based on user input and established design principles. The original and enhanced prototypes were then compared in a study with 15 participants, using TAM and S-TIAS as evaluation frameworks, with follow-up interviews conducted to contextualize the questionnaire responses. The research questions and their conclusions are presented below.

7.1 How can a prototype AI lesson generator for medical educators be extended to better support perceived usefulness, ease of use, and trust?

In this project, the following extensions were implemented: an AI Chat, an Updated Prompt Form, a Lesson Plan, an AI Provenance Banner, and Batch-based Lesson Generation. The evaluation results indicate that these extensions collectively raised perceived usefulness, ease of use, and trust by approximately one point across all constructs.

Out of these features, the AI Chat had the most significant impact. According to the interview data, the AI Chat facilitated iterative lesson revision, which increased users' sense of control of the output. It also allowed them to improve outputs rather than starting over the generation process. Additionally, with the AI Chat, users could make edits in the lesson without knowing how to achieve it manually. The AI Chat thus improved both the perceived usefulness and perceived ease of use.

The Updated Prompt Form affected how some users perceived the ease of use, according to the interview data. Although some noted the new version was easier to understand, the feature was not as frequently mentioned as the AI Chat. Perhaps, this was due to the similarity between the Updated Prompt Form and the original, meaning that the changes, while meaningful to some, were not distinct enough to register as a significant improvement for most.

The Lesson Plan was rarely mentioned in the interviews, indicating that it did not have a meaningful impact on the perceived ease of use. This could potentially be due to users

focusing on the generated lesson, rather than an intermediate planning step, or the Lesson Plan not being perceived as a separate feature in the generation flow.

The AI Provenance Banner did not appear to meaningfully affect trust, despite being grounded in both user requests and established design principles. The improved trust score for the extended prototype appears to have been driven by the improved output quality, which may partly reflect the more detailed lesson outputs enabled by Batch-based Lesson Generation. This suggests that transparency features targeting trust may need to be more prominent to have the intended effect, and that output quality is a stronger trust signal than image provenance disclosure in this context.

7.2 How do medical educators perceive the usefulness, ease of use, and trustworthiness of AI-generated lessons?

Overall, medical educators perceived AI-generated lessons positively across all three dimensions: perceived usefulness, perceived ease of use, and trust. Perceived usefulness was the strongest construct for both prototypes, indicating that educators see meaningful value in a tool that can assist with lesson creation. Perceived ease of use was also rated positively, with most participants finding the generation process accessible, particularly when familiar interaction patterns such as chat-based input were available. Trust was the weakest construct of the three, though still rated positively on average, suggesting that educators approach AI-generated content with cautious confidence rather than outright skepticism.

A notable finding was that both prototypes received an equally poor score on the perceived need to verify generated output, regardless of their differing trust scores. This suggests that the drive to review AI-generated lessons is not primarily a function of trust in the system, but is instead rooted in professional accountability, a sense of personal responsibility for content accuracy, and the educators' need to relate to the material they teach. In medical education in particular, where inaccuracies can have downstream effects on patient safety, human review appears to be regarded as a professional duty rather than an optional step. This finding implies that, even with improvements in AI reliability, educators are unlikely to fully automate lesson creation.

7.3 What factors influence medical educators' perceptions of usefulness, ease of use, and trust in AI-generated lessons?

Several influential factors emerged from the qualitative interview data. The quality and coverage of the underlying knowledge base were the most prominent limiting factors. Multiple participants brought up that an extensive image library was crucial for the tool to be genuinely useful. This highlights the importance of infrastructure quality in tools for AI-lesson generation.

The qualities of the generated output, such as its verbosity, level of detail, and accuracy, also influenced perceptions. Too little detail reduced the usefulness of the tool, while overly verbose output was seen as less suitable for direct use in teaching. This suggests that usefulness depends not only on factual correctness but also on how well the generated content adheres to educators' individual preferences.

Trust was further shaped by transparency: participants reported that knowing the origin of the generated materials instilled greater confidence in the system. Several participants explicitly requested textual references in the generated content, indicating that source attribution is a meaningful trust signal.

7.4 How do feature-level enhancements to the prototype affect medical educators' perceived usefulness, ease of use, and trust?

The enhanced prototype (Prototype A) consistently outperformed the original prototype (Prototype B) by approximately one point across all measured dimensions. According to TAM, perceived usefulness and ease of use are strong predictors of technology acceptance, indicating that the implemented enhancements collectively raised educators' willingness to adopt the tool.

Not all features contributed equally to this outcome. The AI chat was the most impactful feature, according to the interview data. Participants valued the ability to iteratively refine lessons through conversational interaction, as it simplified lesson revision and reduced the need to restart the generation process. The option for freehand prompting was also noted positively by some, offering flexibility. Features such as the lesson plan preview and the AI Provenance Banner, while grounded in design principles, went largely unnoticed by users.

7.5 Implications

The results are an indication that educators are willing to adopt generative AI tools for lesson creation, provided that the underlying data is sufficiently extensive and that the quality of the generated materials is adequate. The tool should be designed for iterative human oversight and human-AI collaboration.

For Sectra, the findings indicate that expanding the image library and implementing textual references in generated content are two of the highest-priority improvements for increasing usefulness and trust. In the interviews, several participants described verifying AI-generated output for reasons unrelated to distrust, such as professional accountability and responsibility for content they would put their name to. This suggests that the perceived need to verify is not driven by distrust alone. For researchers and practitioners in educational AI, this finding is a relevant consideration when designing tools for domains such as medical education.

7.6 Future work

One direction for continued work is to assess how educator perceptions evolve with sustained use of the tool, and in particular, whether the perceived need to verify output decreases over time as familiarity grows, as suggested by participants in this study. Using a multi-rater approach, i.e., involving several researchers when analyzing results, could also strengthen the reliability of the thematic findings by reducing the impact of a single researcher's interpretation.

Future work could also entail expanding the knowledge base to cover a wider range of medical specialties, as the knowledge base in this study was skewed towards radiology and histology. Expanding the knowledge base would also enable a more representative evaluation by including participants from a broader range of medical specialties. Other possible extensions of this research include introducing references to trustworthy sources in the generated text material, implementing textual RAG alongside the existing image-based RAG, and extending the AI provenance feature to cover generated text in addition to images.

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A Interview guide for initial interviews

This appendix presents the interview guide used during the semi-structured interviews described in Section 4.1.3. The field guide served as a flexible framework, guiding the interviews without strictly structuring them.

Introduction

1. Thank you for helping!
2. Consent:
    a) You are participating out of free will and can at any moment stop the interview and leave.
    b) Is it okay if I record the meeting? The records will be available only to me and will be destroyed as soon as I have transcribed and anonymized the data.
    c) Does that sound okay to you?
3. This is an early prototype, and technical issues may occur. If this happens, I have prepared backup materials.
4. I am not testing you; I am testing the tool. There are no right or wrong answers.
5. I am interested in your honest reactions, both positive and negative.

Participant Background

1. Could you describe your role?
2. Do you use lessons in Sectra Education Portal in your role?
3. How do you typically use lessons? How often?

Testing

1. Try generating a lesson for a topic you might realistically teach. It could, for example, match the last lecture you gave.
    a) Radiology example: Benign vs. malignant features of lung nodules
    b) Histopathology example: Normal tissue architecture of the liver
2. Please think aloud as you test the tool.

Questions

1. Tell me about the experience of using the lesson generator. What did you think of it? What was your overall impression?
2. Would you use this lesson as is?
3. How much would you want to change about it? And what?
4. Modifying the lesson — what would you like that process to look like?
5. How would you want to give input to the model?
6. What felt most useful?
7. What would make this genuinely useful in your workflow?
8. What do you think of the output? What is missing or wrong with it?
9. Would you trust this in practice? Why / why not?

Additional Suggested Questions (if time allows)

1. What would stop you from using it?
2. How does this compare to how you currently work?
3. If you could change one thing immediately, what would it be?
4. What was your overall impression?

Wrap-up

1. Is there anything else you think I should know?
2. Thank you so much for helping us — it is deeply appreciated!

B Evaluation form

The following form contains the standardized instructions, tasks, and questionnaire completed by participants during the evaluation, as described in Section 4.3. Eight participants received this version, in which Prototype A (extended) was evaluated before Prototype B (original). The remaining seven participants received a version presenting the prototypes in the opposite order.

Evaluation of AI Lesson Generator

* Required

About the study

Background
This study is part of my master's thesis at Linköping University, conducted in collaboration with Sectra. You will test two different prototypes for AI-assisted lesson creation in Education Portal.

For each prototype, you will receive instructions on how to test it, followed by an evaluation form based on your experience.
This form is divided into two parts:

1. Prototype A:
    - Overview
    - Tasks to complete
    - Evaluation of the prototype

2. Prototype B:
    - Overview
    - Tasks to complete
    - Evaluation of the prototype

You don't have to complete everything in one sitting, but try to fill in the evaluation shortly after testing the prototype while your experience is still fresh.
Important: If you close this form before submitting, your answers will be lost and you'll need to start over.

Data collection and consent

Your participation is voluntary and you may withdraw at any time by emailing **thea.borg@sectra.com**.
If you withdraw, all data associated with you will be deleted upon request, provided it has not already been de-identified to the point where it can no longer be linked to you.
All collected data will be de-identified and used solely for my master's thesis. It will not be shared with anyone outside of this project.

Thank you for participating in this study!

1

What is your name: *

Your name is only used to connect your responses to your follow-up interview and will not appear in any published results. All data will be de-identified after the interview.

2

What level do you teach at? *

- [ ] Upper secondary (Gymnasienivå)
- [ ] Undergraduate (Universitet, grundnivå)
- [ ] Postgraduate (Universitet, avancerad nivå)
- [ ] Residency or specialist training (ST-utbildning)
- [ ] Continuing professional development (Fortbildning)
- [ ] Other

3

What is your area of teaching? *

4

How many lessons have you previously created in Education Portal? *

- ◯ None
- ◯ 1-5
- ◯ 6-15
- ◯ More than 15

Prototype A: Overview

We will now move on to Prototype A. Before you start, here is a brief overview of how the tool works.

Prototype A can be used to generate a lesson using either a **guided form or a freehand prompt**. The system generates a complete lesson based on your input.
Once generated, you can refine the lesson **manually** or using the **AI chat**. You can ask the AI to edit text, adjust questions, restructure content, and more. Each suggestion can be **accepted or discarded**, and you can **undo/redo** AI changes via the AI chat.

In the next section, you will be given instructions on how to test this prototype.

Prototype A: Tasks to complete

Note: As you work, pay attention to accuracy, clarity, and relevance of the generated content, as you will be asked about this in the evaluation.

5

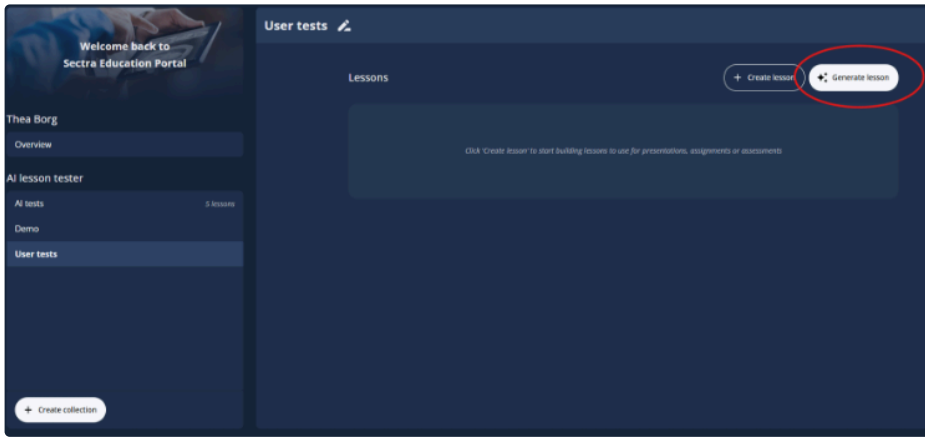
Generate a lesson.

1. Open the following link (keep this form open in a separate tab as you will need it for further instructions):
   https://theas-thesis-va.epsectra-qa.com/home/my-education/?slug=seailt-oe3MCY

2. Press the *Generate lesson* button on the right side of the screen (see image below).

3. Imagine you are preparing a lesson for use in your role. Choose a topic you are familiar with and generate a lesson using either the guided form or the freehand prompt.

You will first be presented with a lesson overview that you can review and give feedback on before generating the full lesson.
Once generated, the lesson will open in edit mode automatically. Take a moment to review the quality and accuracy of the content before moving on.



6

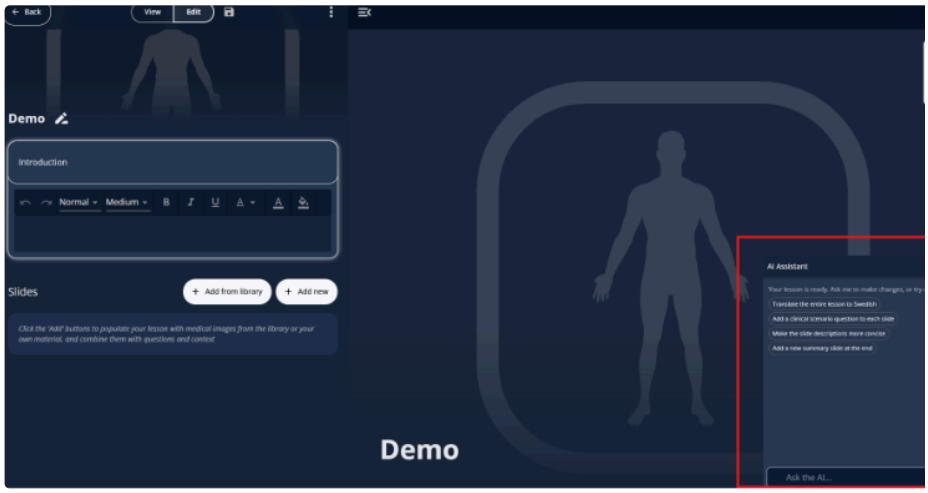
Edit and refine the lesson.

Your next task is now to modify and improve the lesson. First, make sure the lesson is in edit mode.

You can make changes in two ways:
- Use the **AI chat**, which opens when you enter edit mode (see image below).
- Edit the lesson **manually.**

Here are some examples of changes you can try:
- Edit or rewrite text
- Adjust or replace questions
- Change the structure
- Adapt to a different difficulty level

Make sure to try the AI chat at least once.



7

Try a more targeted request

Generate another lesson on a different topic. If your first lesson felt for example felt too general or too short, try being more specific or making this one longer. The goal is to explore the tool further and apply what you learned from your first lesson.

Prototype A: Evaluation

This part of the questionnaire aims to evaluate your experience using Prototype A. It includes statements about usability, usefulness, and your trust in the system. Please indicate how much you agree or disagree with each statement based on your experience. There are no right or wrong answers, your honest feedback is highly valuable.

8

Perceived usefulness *

Please rate each statement on a scale from **Extremely likely (1)** to **Extremely unlikely (7)**.

	Extremely likely	Quite likely	Slightly likely	Neither likely nor unlikely	Slightly unlikely	Quite unlikely	Extremely unlikely
Using Prototype A in my job would enable me to accomplish tasks more quickly.	◯	◯	◯	◯	◯	◯	◯
Using Prototype A would improve my job performance.	◯	◯	◯	◯	◯	◯	◯
Using Prototype A in my job would increase my productivity.	◯	◯	◯	◯	◯	◯	◯
Using Prototype A would enhance my effectiveness on the job.	◯	◯	◯	◯	◯	◯	◯
Using Prototype A would make it easier to do my job.	◯	◯	◯	◯	◯	◯	◯
I would find Prototype A useful in my job.	◯	◯	◯	◯	◯	◯	◯

9

Perceived ease of use *

Please rate each statement on a scale from **Extremely likely (1)** to **Extremely unlikely (7)**.

	Extremely likely	Quite likely	Slightly likely	Neither likely nor unlikely	Slightly unlikely	Quite unlikely	Extremely unlikely
Learning to operate Prototype A would be easy for me.	◯	◯	◯	◯	◯	◯	◯
I would find it easy to get Prototype A to do what I want it to do.	◯	◯	◯	◯	◯	◯	◯
My interaction with Prototype A would be clear and understandable.	◯	◯	◯	◯	◯	◯	◯
I would find Prototype A to be flexible to interact with.	◯	◯	◯	◯	◯	◯	◯
It would be easy for me to become skillful at using Prototype A.	◯	◯	◯	◯	◯	◯	◯
I would find Prototype A easy to use.	◯	◯	◯	◯	◯	◯	◯

10

Trust *

Please rate each statement on a scale from **Strongly agree (1)** to **Strongly disagree (7)**.

	Strongly agree	Agree	Somewhat agree	Neither agree nor disagree	Somewhat disagree	Disagree	Strongly disagree
I can trust Prototype A.	◯	◯	◯	◯	◯	◯	◯
Prototype A is reliable.	◯	◯	◯	◯	◯	◯	◯
I am confident in Prototype A.	◯	◯	◯	◯	◯	◯	◯
I would need to carefully verify Prototype A's output.	◯	◯	◯	◯	◯	◯	◯

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Would you like to share anything else about your experience with Prototype A?

Prototype B: Overview

You are now finished with Prototype A. **Please close that browser tab before continuing.**

You will now move on to Prototype B. Before testing, here is a brief overview of how the tool works.

Prototype B can be used to generate a lesson using a **guided form**. The system generates a complete lesson based on your input.
Once generated, you can refine the lesson **manually**.

In the next section, you will be given instructions on how to test this prototype.

Prototype B: Tasks to complete

Note: As you work, pay attention to accuracy, clarity, and relevance of the generated content, as you will be asked about this in the evaluation.

12

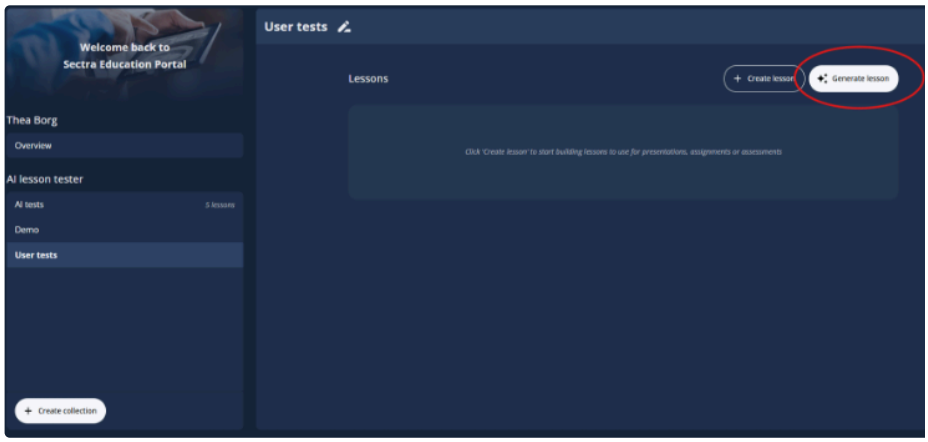
Generate a lesson.

1. Open the following link (keep this form open in a separate tab as you will need it for further instructions):
   https://theas-thesis-vb.epsectra-qa.com/home/my-education/?slug=seailt-JBVHeZ

2. Press the *Generate lesson* button on the right side of the screen (see image below).

3. Imagine you are preparing a lesson for use in your role. Choose a topic you are familiar with and generate a lesson using either the guided form or the freehand prompt.

Once generated, the lesson will open in edit mode automatically. Take a moment to review the quality and accuracy of the content before moving on.



13

Edit and refine the lesson.

Your next task is now to modify and improve the lesson. First, make sure the lesson is in edit mode.

You can make changes by editing the lesson manually.

Here are some examples of changes you can try:
- Edit or rewrite text
- Adjust or replace questions
- Change the structure
- Adapt to a different difficulty level

14

Try a more targeted request

Generate another lesson on a different topic. If your first lesson felt for example felt too general or too short, try being more specific or making this one longer. The goal is to explore the tool further and apply what you learned from your first lesson.

Prototype B: Evaluation

This part of the questionnaire aims to evaluate your experience using Prototype B. It includes statements about usability, usefulness, and your trust in the system. Please indicate how much you agree or disagree with each statement based on your experience. There are no right or wrong answers, your honest feedback is highly valuable.

15

Perceived usefulness *

Please rate each statement on a scale from **Extremely likely (1)** to **Extremely unlikely (7)**.

	Extremely likely	Quite likely	Slightly likely	Neither likely nor unlikely	Slightly unlikely	Quite unlikely	Extremely unlikely
Using Prototype B in my job would enable me to accomplish tasks more quickly.	◯	◯	◯	◯	◯	◯	◯
Using Prototype B would improve my job performance.	◯	◯	◯	◯	◯	◯	◯
Using Prototype B in my job would increase my productivity.	◯	◯	◯	◯	◯	◯	◯
Using Prototype B would enhance my effectiveness on the job.	◯	◯	◯	◯	◯	◯	◯
Using Prototype B would make it easier to do my job.	◯	◯	◯	◯	◯	◯	◯
I would find Prototype B useful in my job.	◯	◯	◯	◯	◯	◯	◯

16

Perceived ease of use *

Please rate each statement on a scale from **Extremely likely (1)** to **Extremely unlikely (7)**.

	Extremely likely	Quite likely	Slightly likely	Neither likely nor unlikely	Slightly unlikely	Quite unlikely	Extremely unlikely
Learning to operate Prototype B would be easy for me.	◯	◯	◯	◯	◯	◯	◯
I would find it easy to get Prototype B to do what I want it to do.	◯	◯	◯	◯	◯	◯	◯
My interaction with Prototype B would be clear and understandable.	◯	◯	◯	◯	◯	◯	◯
I would find Prototype B to be flexible to interact with.	◯	◯	◯	◯	◯	◯	◯
It would be easy for me to become skillful at using Prototype B.	◯	◯	◯	◯	◯	◯	◯
I would find Prototype B easy to use.	◯	◯	◯	◯	◯	◯	◯

17

Trust *

Please rate each statement on a scale from **Strongly agree (1)** to **Strongly disagree (7)**.

	Strongly agree	Agree	Somewhat agree	Neither agree nor disagree	Somewhat disagree	Disagree	Strongly disagree
I can trust Prototype B.	◯	◯	◯	◯	◯	◯	◯
Prototype B is reliable.	◯	◯	◯	◯	◯	◯	◯
I am confident in Prototype B.	◯	◯	◯	◯	◯	◯	◯
I would need to carefully verify Prototype B's output.	◯	◯	◯	◯	◯	◯	◯

18

Would you like to share anything else about your experience with Prototype B?

Final thoughts

Thank you for testing both prototypes! I am very grateful for your participation.

Please make sure you have booked a time for your follow-up interview here: https://doodle.com/meeting/participate/id/dPK9q8yb

You can reach me at thea.borg@sectra.com if you have any questions after submitting this form.

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If you have any additional comments, feedback, or questions, feel free to share them below.

This content is neither created nor endorsed by Microsoft. The data you submit will be sent to the form owner.

Microsoft Forms

C Interview guide for final interviews

This appendix presents the interview guide used during the semi-structured interviews conducted as part of the evaluation phase, as described in Section 4.3.5. The guide served as a flexible framework, steering the conversation without strictly structuring it.

Introduction

1. Thank you for helping!
2. I am interested in your honest thoughts and opinions, both positive and negative. There are no right or wrong answers.
3. Consent:
    a) Is it okay if I record this interview? The recording will only be used for analysis, your data will be anonymous, and the recording will be deleted after the thesis is completed.
    b) You are participating out of your own free will and can leave at any time without giving a reason.

Participant Background

1. Could you describe your role?
2. Do you use lessons in Sectra Education Portal in your role?
    a) If so, how?

General Questions

1. Did you encounter any bugs?
2. Tell me about your overall experience. What stood out?
3. How did the two prototypes compare?

a) Which did you prefer and why?

Ease of Use

1. How did you find the process of generating a lesson with Prototype A? What about Prototype B?
2. How was the editing experience?
    a) What did you think about the AI chat?
3. What was confusing or frustrating?

Usefulness

1. How would you feel about using either prototype in your day-to-day work?
2. Could you see this saving you time in your role?
3. What would need to change for this to be truly useful to you?

Trust

1. What did you think about the quality of the generated lessons, in terms of accuracy, tone, and structure?
    a) Did you notice any differences between A and B?
2. What made you trust or distrust the output?
3. Did you feel the need to verify the content?
    a) If yes: How would you go about that?
4. Would you feel comfortable using a generated lesson with students without significant manual review?

Wrap-up

- Is there anything else you would like to share?
- Do you have any questions for me?
- Thank you!